# 2019 International Dictyostelium Meeting
# Ann Arbor, MI 48109, USA


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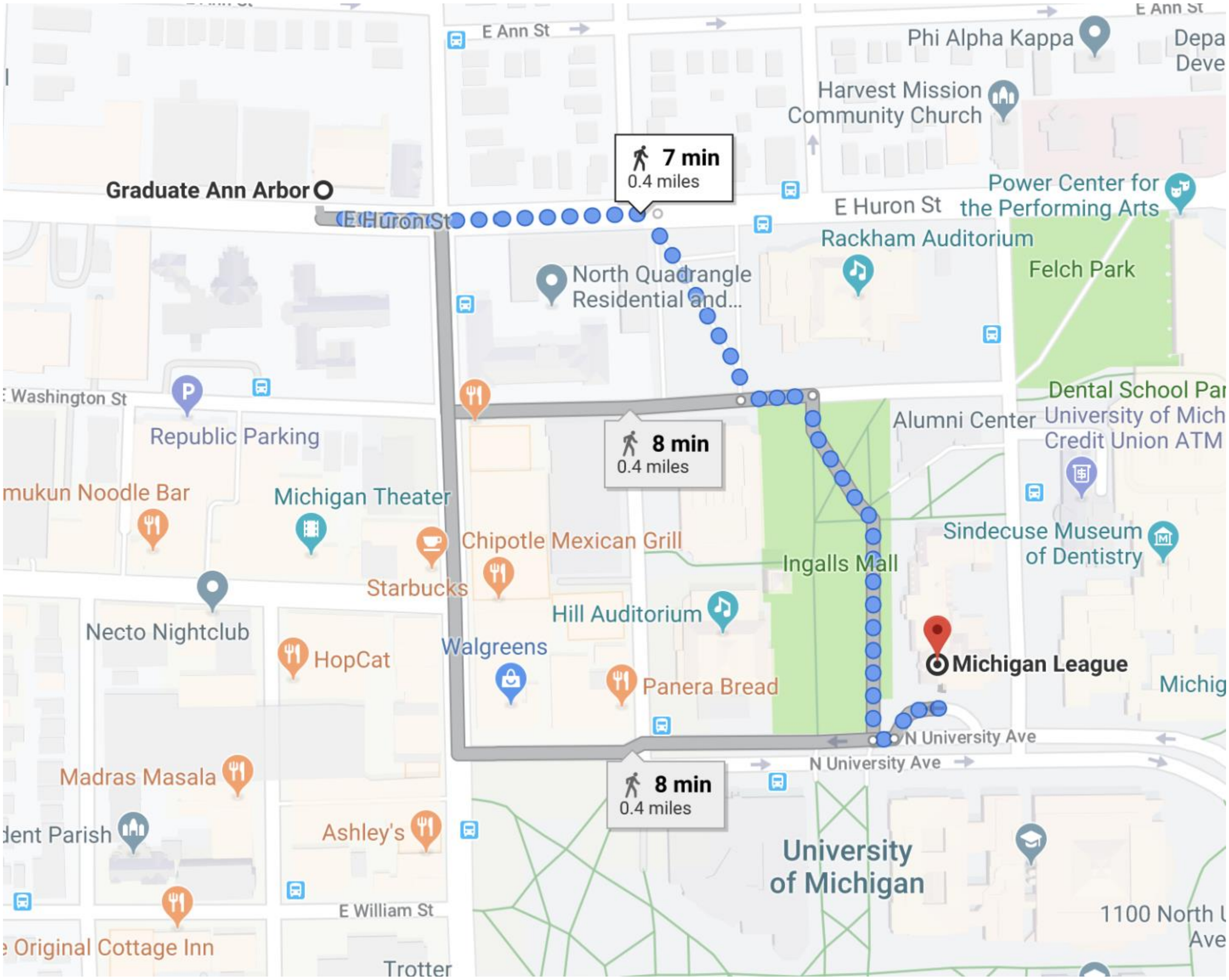

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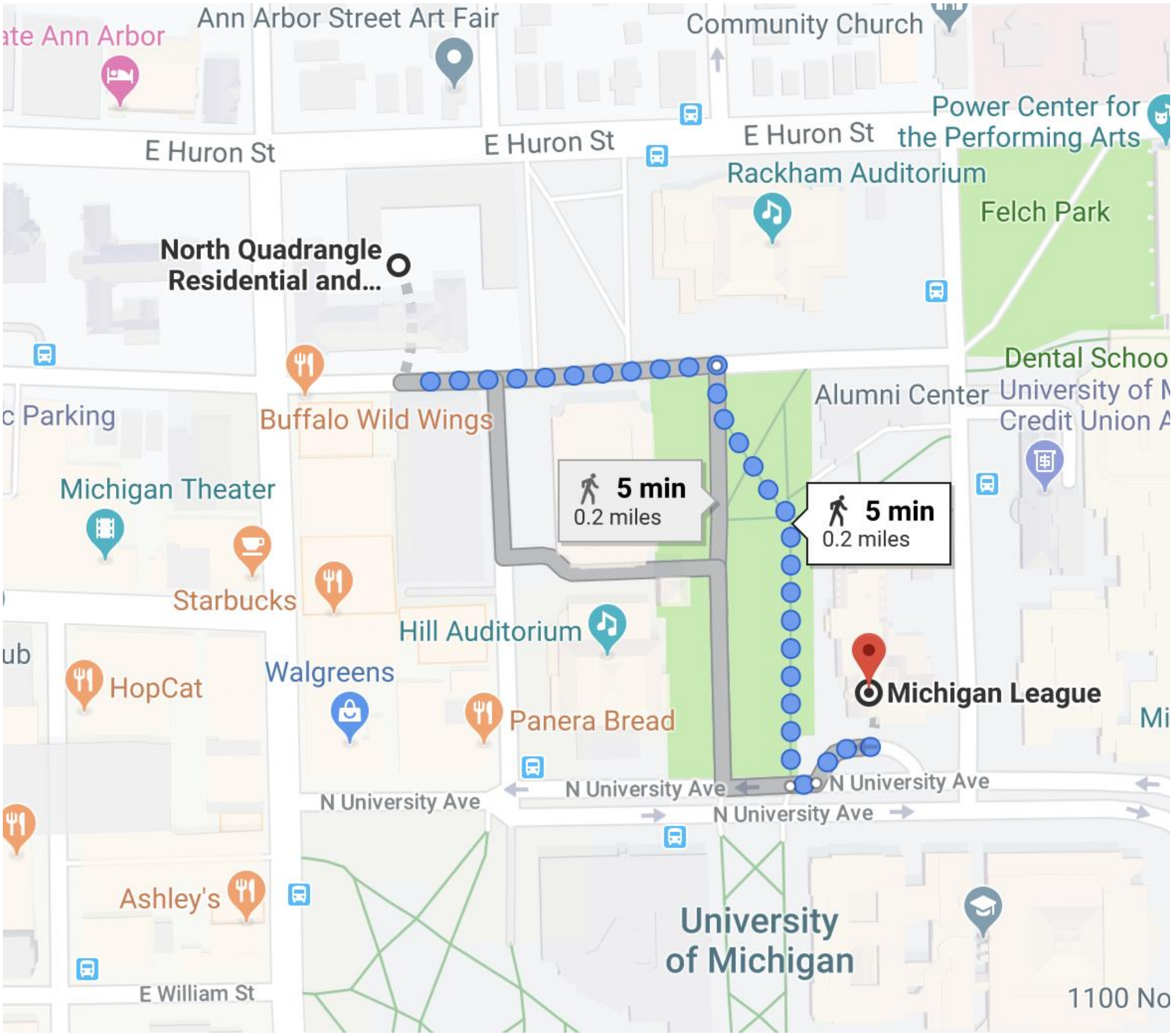

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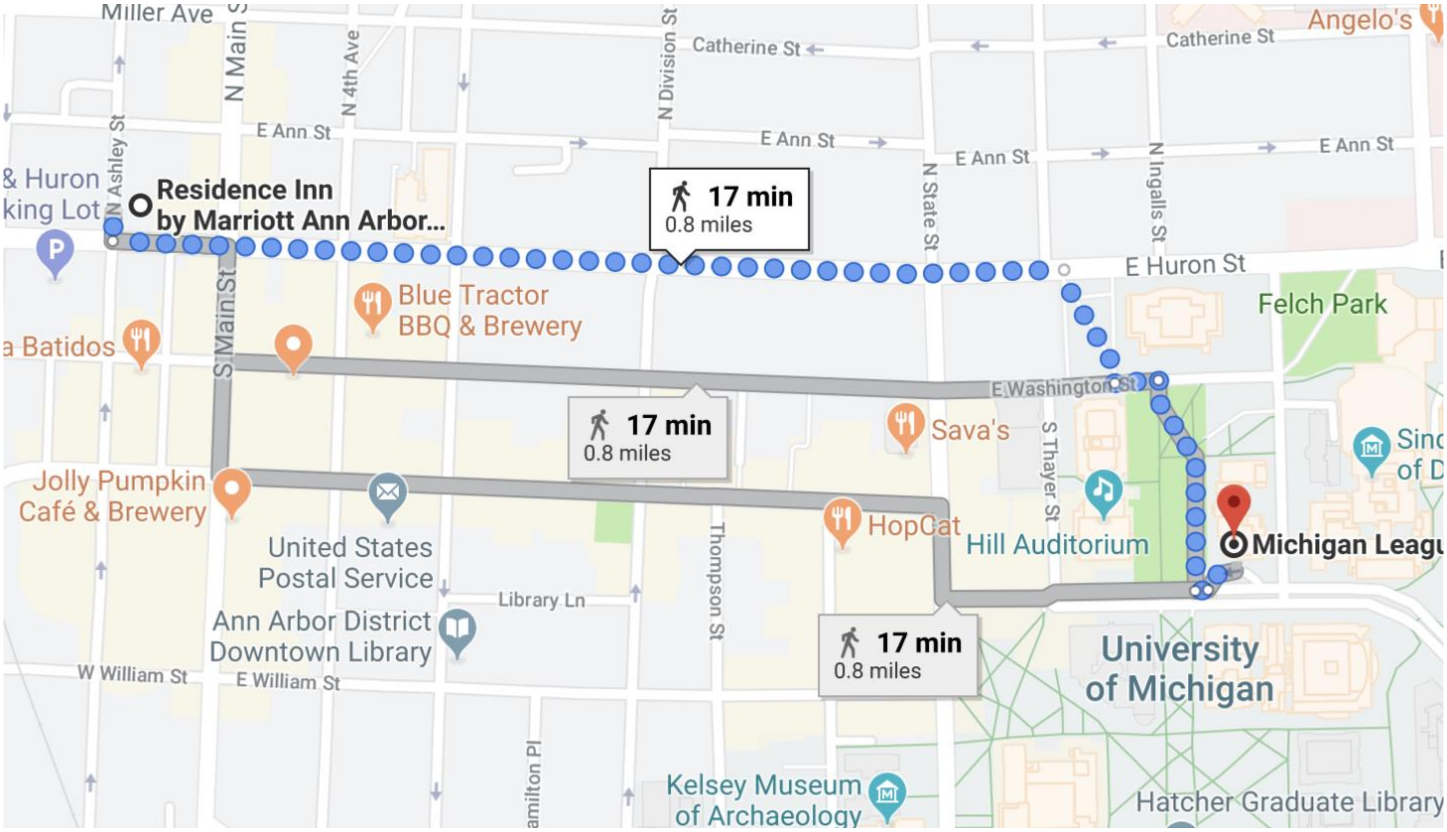

# Map of the 2nd floor of the Michigan League

**MICHIGAN LEAGUE**

**DICTY CONFERENCE 2019**

**Registration:** Concourse
**Meeting Location:** Hussey
**Meals & Posters:** Ballroom


# Michigan League Contact Information:

MI League Address: 911 North University Ann Arbor, MI 48109
Phone Number: Information Desk: 734-647-5343

# 2019 International *Dictyostelium* Meeting, Ann Arbor, MI

## Sunday, August 4th

**2:00 – 6:00 Registration – Michigan League Concourse**

**6:00 – 7:00 Keynote Lecture**
**Cell migration from a heterotrimeric G protein biologist's perspective: it all starts here!**
Alan Smrcka, Ph.D.
Benedict R. Lucchesi Collegiate Professor of Cardiovascular Pharmacology
Department of Pharmacology
University of Michigan Medical School
1150 W. Medical Center Drive
MSRB II, Room 5560C
Ann Arbor, MI 48109

**7:00 – 10:00 Reception/Mixer**

## Monday, August 5th

**7:30 – 9:00 Breakfast**

**Session 1: Cell Biology 1 (9:00 – 10:40)**
**Chair: Rob Huber, Trent University**

**9:00 – 9:25**
**1. Cell-Autonomous and Non-Autonomous Functions for Growth and Density-Dependent Development of *Dictyostelium* Regulated by Ectodomain Shedding**
Fu-Sheng Chang, Pundrik Jaiswal, Netra Pal Meena, Joseph Brzostowski, and Alan R. Kimmel

**9:25 – 9:50**
**2. Profiling of cytokinin levels during the *Dictyostelium* life cycle and their effects on cell proliferation and spore germination**
Megan M. Aoki, Craig Brunetti, Robert J. Huber, & R. J. Neil Emery

**9:50 – 10:15**
**3. Proteostatic mechanisms of Dictyostelium discoideum**
Stephanie Santarriaga, Holly Haver, K Matthew Scaglione

**10:15 – 10:40**
**4. CpnA has a role in contractile vacuole function and postlysosome maturation**
Elise Wight, Amber Ide, and Cynthia Damer

**10:40 – 11:00 Break**

**Session 2: Cell Biology 2 (11:00 – 12:40)**
**Chair: Chris Janetopoulos, University of the Sciences**

**11:00 – 11:25**
**5. A Model for Bleb Nucleation in *Dictyostelium discoideum***
E.O. Asante-Asamani, Zully Santiago, Devarshi Rawal, John Loustau, and Derrick Brazill

**11:25 – 11:50**
**6. IqgC is an atypical IQGAP-related protein that attenuates Ras signaling during large-scale endocytosis**
Maja Marinović, Lucija Mijanović, Marko Šoštar, Matej Vizovišek, Alexander Junemann, Marko Fonović, Boris Turk, Igor Weber, Jan Faix and Vedrana Filić

**11:50 – 12:15**
**7. Microtubule Array Dynamics in Dictyostelium**
Michael P. Koonce, Jacob Odell, and Irina Tikhonenko

**12:15 – 12:40**
**8. Glycoregulation in the nucleus and cytoplasm of *Dictyostelium* and other protists**
Christopher M. West[1], Hanke van der Wel, Ana Maria Garcia Iguaran, Giulia Bandini, John Samuelson

**12:40 – 2:00 Lunch**

**Session 3: Development and Evolution (2:00 – 3:40)**
**Chair: Pascale Charest, University of Arizona**

**2:00 – 2:25**
**9. Role of *PEPC* gene in prespore lineage choice and differentiation in *D.discoideum***

Kenichi Abe, Satoshi Kuwana, Kazuteru Taoka, Hidenori Hashimura, Satoshi Sawai, Masashi Fukuzawa

**2:25 – 2:50**
**10. Cellular plasticity in *Dictyostelium***
John Nichols, Vlatka Antolovic, Tchern Lenn, Jacob Reich and Jonathan R. Chubb
MRC Laboratory for Molecular Cell Biology and Department of Cell and Developmental Biology, University

**2:50 – 3:15**
**11. Volatile Signals Synchronize Development in Social Amoebae**
Amanda Webb, Mariko Kurasawa-Katoh, Rafael Rosengarten, Xinlu Chen, Feng Chen, and Gad Shaulsky

**3:15-3:40**

**12. Eco-evolutionary significance of 'loners'**
Fernando W. Rossine, Ricardo Martinez-Garcia, Allyson E. Sgro, Thomas Gregor, Corina E. Tarnita

**3:40 – 4:00 BREAK**

**Session 4: *Dictyostelium* as a disease model (4:00 – 5:40)**
**Chair: Kari Naylor, University of Central Arkansas**

**4:00 – 4:25**
**13. Recent insights into Cln5 and Mfsd8 function in *Dictyostelium***
Robert J. Huber

**4:25 – 4:50**
**14. GGPP Depletion Selectively Kills Cells Lacking PTEN by Blocking Motility and Macropinocytosis**
Zhihua Jiao, Huaqing Cai, Yu Long, Orit Katarina Sirka, Andrew Ewald, and Peter N Devreotes

**4:50 – 5:15**
**15. Autophagy is upregulated in a *Dictyostelium* model of CLN5 disease**
Meagan D, McLaren**,** William Kim, Sabateeshan Mathavarajah, Robert J. Huber

**5:15 – 5:40**
**16. *Dictyostelium discoideum* as a model for Parkinson's disease?**
Ethan Chernivec, Jacie Cooper and Kari Naylor

**5:40 – 6:00 Break**

**6:00 – 7:30 Dinner**

**7:30 – 10:00 Poster Session 1**

## Tuesday, August 6th

**7:30 – 9:00 Breakfast**

**Session 5: Host Microbe Interactions (9:00 – 10:40)**
**Chair: Tera Levin, Fred Hutchinson Cancer Research Center**

**9:00 – 9:25**
**17. Polyphosphate inhibits the killing of ingested bacteria in *Dictyostelium* and macrophages**
Ramesh Rijal, Morgan R. Smith and Richard H. Gomer

**9:25 – 9:50**
**18. Discoidin I regulation of bacterial carriage during the growth to development transition in *Dictyostelium discoideum*.**
Christopher Dinh and Adam Kuspa

**9:50 – 10:15**
**19. How amoebae locate and eat bacteria-- G-protein-coupled receptor mediated cell migration and bacterial engulfment**
Miao Pan and Tian Jin

**10:15 – 10:40**
**20. AlyL: a *D. discoideum* putative lysozyme with an antibacterial activity**
Tania Jauslin, Otmane Lamrabet, Pierre Cosson

**10:40 – 11:00 BREAK**

**Session 6: Host Microbe Interactions 2 (11:00 – 12:15)**
**Chair: Marco Tarantola, Max Planck Institute for Dynamics and Self-Organization**

**11:00 – 11:25**
**21. "Evolutionary battles for iron between Dicty and Legionella**
Tera Levin, Killian Campbell, and Harmit Malik

**11:25 – 11:50**
**22. Dual RNA-seq of naïve *D. discoideum* and their *Burkholderia* endosymbionts hints at different evolutionary histories**
Suegene Noh[1], Susanne DiSalvo[2], David Queller[3], Joan Strassmann[3]

**11:50 – 12:15**
**23. Fine tuning of the Anionic Membrane Lipids PI(4,5)P2 and Phosphatidylserine Establishes Polarized Morphologies and Regulates Cell Migration**
Mariam Beshay, Adelle Schade, Andy Ring, Nada Bawazir, and Chris Janetopoulos

**12:15 – Pick up boxed lunch/ Free Time/ Ford Museum/ Canoeing**

**6:00 – 7:30 Dinner**

**7:30 – 10:00 Poster Session 2**

## Wednesday, August 7th

**7:30 – 9:00 Breakfast**

**Session 7: Workshops (9:00 – 10:30)**

**9:00 – 9:30**
**24. GoldenBraid cloning for synthetic biological applications in *Dictyostelium***
Peter Kundert, Alejandro Sarrion-Perdigones, Yezabel Gonzalez, Mariko Kurasawa, Shigenori Hirose, Peter Lehmann, Chris Dinh, Timothy Farinholt, Adam Kuspa, Koen Venken, and Gad Shaulsky

**9:30 – 10:00**
**25. dictyBase Version 2 - work in progress**
Petra Fey, Eric Hartline, Rex L Chisholm, and Siddhartha Basu

**10:00 – 10:30 Grant Funding**
Richard Gomer and Carole Parent

**10:30 – 11:00 BREAK**

**Session 8: Chemotaxis and Signal Transduction 1 (11:00 – 12:40)**
**Chair: Allyson Sgro, Boston University**

**11:00 – 11:25**
**26. Phosphorylated Rho-GDP Directly Activates mTORC2 Kinase Toward AKT Through Dimerization with Ras-GTP to Regulate Cell Migration**
Miho Iijima

**11:25 – 11:50**
**27. Adhesion strategies of Dictyostelium discoideum – a force spectroscopy study**
Nadine Kamprad, Hannes Witt, Marcel Schroeder, Christian Titus Kreis, Oliver Baeumchen, Andreas Janshoff and Marco Tarantola

**11:50 – 12:15**
**28. The Atypical MAP Kinase ErkB Transmits Distinct Chemotactic Signals through a Core Signaling Module**
John M.E. Nichols, Peggy Paschke, Sew Peak-Chew, Thomas D. Williams, Luke Tweedy, Mark Skehel, Elaine Stephens, Jonathan R. Chubb, Robert R. Kay

**12:15 – 12:40**
**29. Negative Surface Charge Defines the State of Cell Cortex and Regulates Excitable Dynamics**
Tatsat Banerjee, Debojyoti Biswas, Dhiman Sankar Pal, Yuchuan Miao, Pablo A Iglesias, Peter N Devreotes

**12:40 – 2:00 Lunch**

**Session 9: Chemotaxis and Signal Transduction 2 (2:00 – 3:40)**
**Chair: Yulia Artemenko, SUNY Oswego**

**2:00 – 2:25**
**30. The torsinA homologue tsin is required for the multicellular development of *Dictyostelium discoideum***
Saunders CA, Erickson JR, Woolums BM, Bauer H, Titus MA, and Luxton GWG

**2:25 – 2:50**

**31. Proteomic analysis of chemoattractant-induced signaling dynamics**
Yihong Yang, Xiaoting Chao, Dong Li and Huaqing Cai

**2:50 – 3:15**
**32. Roles of typical and atypical MAP kinases in *Dictyostelium***
Jeff Hadwiger, Huaqing Cai, and David Schwebs

**3:15 – 3:40**
**33. Identifying the knobs used to modulate collective signaling in *Dictyostelium***
Allyson E. Sgro

**3:40 – 4:00 Break**

**Session 10: Chemotaxis and Signal Transduction 3 (4:00 – 5:40)**
**Chair: Suegene Noh, Colby College**

**4:00 – 4:25**
**34. Reduced oxygen availability triggers aerotactic migration in Dictyostelium**
Cochet-Escartin O, Rieu J.P and Anjard C

**4:25 – 4:50**
**35. TalA, MhcA, and SCAR regulate blebs in response to environmental compression in *Dictyostelium discoideum***
Zully Santiago and Derrick Brazill

**4:50 – 5:15**
**36. How Does a Ras Inhibitor Control the Gradient-Sensing Dynamics and Chemotaxis in *Dictyostelium discoideum?***
Xuehua Xu and Tian Jin

**5:15 – 5:40**
**37. Exosomes as key regulators of signal relay during chemotaxis**
Carole A. Parent

**5:40– 6:00 Break**

**6:00 – 10:00 Banquet/Music**

## Thursday, August 8th

**Depart**

# Abstracts

## Talks

### 1. Cell-Autonomous and Non-Autonomous Functions for Growth and Density-Dependent Development of *Dictyostelium* Regulated by Ectodomain Shedding

Fu-Sheng Chang*, Pundrik Jaiswal*, Netra Pal Meena*, Joseph Brzostowski*, ^, and Alan R. Kimmel*

* Laboratory of Cellular and Developmental Biology
National Institute of Diabetes and Digestive and Kidney Diseases
The National Institutes of Health, Bethesda, MD 20892, USA

and

^ Laboratory of Immunogenetics Twinbrook Imaging Facility
National Institute of Allergy and Infectious Diseases
The National Institutes of Health, Rockville, MD 20852, USA

Cell-cell interactions and response are enhanced by increased cell density. We were interested to identify novel secreted proteins that accumulate in parallel to the collective local cell population and that can direct developmental decisions and we hypothesized that novel secreted proteins may serve as density-sensing factors to promote multi-cell developmental fate decisions at specific cell-density thresholds. We show that multi-cell developmental aggregation in *Dictyostelium* is lost upon minimal (2-fold) reduction in local cell density. Remarkably, aggregation response at non-permissive cell densities is rescued by addition of conditioned media from high density, developmentally competent cells. Using rescued aggregation of low-density cells for assay, we purified a single, 150 kDa extra-cellular protein with density aggregation activity. MS/MS peptide sequence analysis identified the gene sequence, and cells that overexpress p150 accumulate higher levels of the Developmental Promoting Factor (DPF) activity than parental cells and aggregate at lower cell densities; cells deficient for *DPF* lack density-dependent aggregation activity and require higher cell density for cell aggregation, compared to WT. The density aggregation activity co-purifies with tagged versions of DPF and tag-affinity purified DPF possesses density aggregation activity. In mixed developmental studies with WT cells, cells overexpressing DPF define centers for multi-cell aggregation and determine cell-fate choice during *Dictyostelium* cytodifferentiation. Finally, we show that DPF is synthesized as a larger precursor, single-pass transmembrane protein that is released by proteolytic cleavage and ectodomain shedding and that the TM/cytoplasmic domain of DPF possesses cell autonomous activity for cell-substratum adhesion and for cellular growth.

## 2. Profiling of cytokinin levels during the *Dictyostelium* life cycle and their effects on cell proliferation and spore germination

Megan M. Aoki, Craig Brunetti, Robert J. Huber, & R. J. Neil Emery
*Environmental & Life Sciences Graduate Program, Trent University*
*Department of Biology, Trent University*

Cytokinins (CKs) encompass a family of evolutionarily conserved growth regulating hormones. While CKs are well-characterized in plant systems, these $N^6$ adenine derivatives are found in a variety of organisms beyond plants, including bacteria, fungi, mammals, and the social amoeba, *Dictyostelium discoideum*. Within *Dicytostelium*, CKs have only been studied in the late developmental stages of the life cycle, whereby they promote spore encapsulation and dormancy. In this study, we used ultra high-performance liquid chromatography-positive electrospray ionization-high resolution tandem mass spectrometry (UHPLC-(ESI+)-HRMS/MS) to profile CKs throughout the *Dictyostelium* life cycle: growth, aggregation, mound, slug, fruiting body, and germination. Comprehensive profiling revealed that *Dictyostelium* produces 6 CK different types in varying abundance across the sampled life cycle stages: *cis*-Zeatin (*c*Z), discadenine (DA), $N^6$-isopentenyladenine (iP), $N^6$-isopentenyladenine-9-riboside (iPR), $N^6$-isopentenyladenine-9-riboside-5' phosphate (iPRP), and 2-methylthio-$N^6$-isopentenyladenine (2MeSiP). Interestingly, iP-type CKs were the most dominant CK analytes detected in the growth and aggregation life cycle stages and these levels increased during aggregation. Exogenous treatment of cells with various CK types revealed that iP was the only CK to affect the proliferation of AX3 cells in culture. In support of previous studies, metabolomics data revealed that discadenine (DA) is one of the most significantly upregulated small molecules during *Dictyostelium* development, and our data indicates that CK levels are highest during germination. While much remains to be explored in *Dictyostelium*, this research offers new insight into the novel production of CK forms by this organism and perhaps greater roles of CKs beyond the fruiting body and germination life cycle stages.

### 3. Proteostatic mechanisms of *Dictyostelium discoideum*

Stephanie Santarriaga[1], Holly Haver[2], K Matthew Scaglione[2,3,4]
[1]*Department of Biochemistry, Medical College of Wisconsin, Milwaukee, Wisconsin 53226, USA*
[2]*Department of Molecular Genetics and Microbiology, Duke University School of Medicine, Durham, North Carolina 27110, USA*
[3]*Department of Neurology, Duke University School of Medicine, Durham, North Carolina 27110, USA*
[4]*Duke Center for Neurodegeneration and Neurotherpeutics, Duke University School of Medicine, Durham, North Carolina 27110, USA*

Proteopathies are a class of at least 71 diseases characterized by the accumulation of protein aggregates. Protein aggregates are caused by an imbalance in protein homeostasis resulting in the accumulation of misfolded proteins. One class of neurodegenerative diseases, the polyglutamine diseases, is cause by a polyglutamine expansion in the coding region of specific genes. In these diseases the polyglutamine-expanded proteins aggregate in cells and induce toxicity. A such one major question in biomedical research is: How do cells recognize and deal with misfolded proteins? Unlike other model organism *Dictyostelium discoideum* normally expresses proteins with long polyglutamine tracts and as such represent a proteostatic outlier that contains vast numbers of proteins that are expected to be aggregation prone. Here we have investigated how *Dictyostelium* deals with these aggregation-prone proteins and will discuss proteostatic mechanisms Dictyostelium use to suppress protein aggregation.

## 4. CpnA has a role in contractile vacuole function and postlysosome maturation

Elise Wight, Amber Ide, and Cynthia Damer
Central Michigan University

Copines make up a family of cytosolic proteins that associate with membranes in a calcium-dependent manner and are found in many eukaryotic organisms. The presence of two C2 domains suggests copines may have a role in membrane trafficking. *Dictyostelium discoideum* has six copine genes (*cpnA-F*) and cells lacking *cpnA* have been shown to have defects in cytokinesis, chemotaxis, adhesion, and development. GFP-tagged CpnA was shown to associate with the plasma membrane, endosomes, lysosomes, phagosomes, and contractile vacuoles. Here, we use *cpnA-* cells to investigate the role of CpnA in contractile vacuole function and endocytosis. When placed in water, *cpnA-* cells made unusually large contractile vacuoles that took longer to expel. Visualization of contractile vacuoles with the marker protein GFP-dajumin indicated that the contractile vacuoles of *cpnA-* cells made fewer, larger, more persistent vacuoles that began refilling before complete emptying. In endocytosis assays, *cpnA-* cells took up small fluorescent beads by macropinocytosis at rates similar to parental cells. However, at the later timepoints, *cpnA-* cells had less fluorescence and the beads were found in smaller endolysosomal organelles. In a feed-chase experiment, cells were fed FITC- and TRITC-labeled dextran to distinguish neutral postlysosomes from acidic endosomes and lysosomes. Postlysosomes appeared sooner in *cpnA-* cells, did not become as large, and disappeared at a faster rate. p80 antibody staining of postlysosomes also indicated that *cpnA-* cells have smaller postlysosomes. These results suggest that CpnA is involved the regulation of contractile vacuole size and expulsion, and the maturation, size, and exocytosis of postlysosomes.

## 5. A Model for Bleb Nucleation in *Dictyostelium discoideum*

E.O. Asante-Asamani[1,2], Zully Santiago[1], Devarshi Rawal[2], John Loustau[2], and Derrick Brazill[1]

1. Hunter College, CUNY, Department of Biological Sciences
2. Hunter College, CUNY, Department of Mathematical Sciences

*D. discoideum* cells, when moving under compressive forces such as experienced in a mound or pseudoplasmodium, detach local regions of their plasma membrane from the underlying cortex. These blister like pressure driven protrusions are referred to as blebs. A similar motility structure has been observed in white blood cells and cancer cells invading different tissues. The precise mechanism by which cells generate blebs and coordinate their movement using these protrusions is unknown. As a critical step towards a comprehensive understanding of bleb-based motility, we are interested in how cells select locations on their membrane to form blebs. We postulate that blebs form in locations of maximum boundary stress, where myosin II gets recruited to the cortex and eventually helps to rupture it. To test our hypothesis, we have developed a membrane energy functional that captures the underlying forces due to cell geometry and local activity of myosin II. Predictions of bleb nucleation sites using the energy functional are compared to observed nucleation sites from a region of interest analysis of the blebbing region. Our mathematical model accurately predicts 96.8% of observed nucleation sites. By examining the different components of the mathematical model, which represent factors influencing bleb nucleation, we found membrane tension and local hydrostatic pressure to be most significant. In addition, we provide for the first time evidence of cortex gap formation prior to blebbing. We show, in support of our model, that myosin II clusters form at the front of the cell prior to bleb nucleation.

## 6. IqgC is an atypical IQGAP-related protein that attenuates Ras signaling during large-scale endocytosis

Maja Marinović[1], Lucija Mijanović[1], Marko Šoštar[1], Matej Vizovišek[2], Alexander Junemann[3], Marko Fonović[2,4], Boris Turk[2,4,5], Igor Weber[1], Jan Faix[3] and Vedrana Filić[1]

[1]*Division of Molecular Biology, Ruđer Bošković Institute, 10000 Zagreb, Croatia*
[2]*Department of Biochemistry and Molecular and Structural Biology, Jožef Stefan Institute, 1000 Ljubljana, Slovenia*
[3]*Institute for Biophysical Chemistry, Hannover Medical School, 30625 Hannover, Germany*
[4]*Centre of Excellence for Integrated Approaches in Chemistry and Biology of Proteins, Jamova 39, SI-1000 Ljubljana, Slovenia*
[5]*Faculty of Chemistry and Chemical Technology, University of Ljubljana, Večna pot 113, SI-1000 Ljubljana, Slovenia*

Ras proteins are important regulators of bulk fluid uptake (macropinocytosis) and large particles uptake (phagocytosis), collectively named large-scale endocytosis. Ras promotes actin- driven membrane ruffling and cup formation, but its activity has to be precisely regulated. While guanine nucleotide exchange factors are necessary for activation of Ras GTPases, GTPase activating proteins (GAPs) are crucial for their timely inactivation. So far, the only RasGAP known to be responsible for Ras inactivation during large-scale endocytosis is NF1, a *Dictyostelium* orthologue of human RasGAP neurofibromin 1, identified in wild-type *Dictyostelium* strains. However, gene encoding NF1 is largely deleted in axenic strains and cannot account for suppression of Ras signaling on macropinosomes and phagosomes. We discovered that another protein with RasGAP activity, IqgC, attenuates Ras signaling during large-scale endocytosis in axenic cells. Although classified as an IQGAP-related protein, IqgC appears to be an atypical family member. Unlike IQGAPs, IqgC can bind Ras GTPases and has conserved amino acid residues critical for catalytic GAP activity. Interaction studies identified RasG as its endogenous binding partner, whereas biochemical assay confirmed its GAP activity toward this GTPase. Localization data showed prominent recruitment of IqgC to nascent macropinosomes, where it colocalizes with active Ras, and to a lesser extent to phagosomes. Consistently, functional assays with *iqgC*- and IqgC-overexpressing cells demonstrated negative regulatory role of this GAP in both types of large-scale endocytosis. Specifically, we demonstrated that this RasG-specific GAP restrains the size of the macropinosomes and thus suppresses bulk fluid uptake in axenic laboratory strain.

## 7. Microtubule Array Dynamics in Dictyostelium

Michael P. Koonce, Jacob Odell, and Irina Tikhonenko
Wadsworth Center, NYS Department of Health. Albany, NY 12201-0509

Interphase microtubule (MT) arrays in most animal cells extend radially from the centrosome and display a self-centering activity in the cytoplasm due to forces acting at the distal MT ends. Numerous studies have highlighted the dynamics of individual arrays, but relatively few works address the behavior of multiple arrays that coexist in a common cytoplasm. In multinucleated *D. discoideum* cells, each centrosome organizes a radial MT array; a striking feature of these cells is that these arrays remain separate from one another. This property offers an opportunity to understand how multiple arrays interact and reveal mechanism(s) responsible for their positioning. Using a laser microbeam to eliminate one of the two centrosomes in binucleate cells, we show that the unaltered MT array is rapidly repositioned at the cell center. This result demonstrates that each MT array is constantly subject to centering forces and infers there must be a mechanism to balance the positions of multiple arrays. At least two different mechanisms have been described to explain such a balance. The first, rooted in the mitotic literature, is that through actions of motors and crosslinkers, MT arrays of opposite polarity actively engage one another and form a physical barrier that limits interdigitation. The second idea is that simple pushing from MTs or motor proteins can act to set distances. These are not mutually exclusive ideas and do not rule out other contributing factors. Since we have previously shown that the radial arrangement in *D. discoideum* is sensitive to MT motor and crosslinker perturbations, we also address the actions of three kinesins and a crosslinking MAP in this study. Our results suggest that a third mechanism, one involving MT exclusion, contributes to an imbalance of cortical forces that in turn fosters aster separation. Understanding the forces acting upon the MT array is important as it lends insight into the fundamental organization of the cytoplasm. Our work is supported in part by the National Science Foundation.

**8. Glycoregulation in the nucleus and cytoplasm of *Dictyostelium* and other protists**

Christopher M. West[1,2], Hanke van der Wel[1], Ana Maria Garcia Iguaran, Giulia Bandini[3], John Samuelson[3]

[1]Department of Biochemistry & Molecular Biology, [2]Center for Tropical and Emerging Global Diseases, University of Georgia, Athens, GA 30602; [3]Department of Molecular and Cell Biology, Boston University Goldman School of Dental Medicine, Boston, MA 02118 USA

Glycosylation is best known on proteins that traverse the secretory pathway, but select nuclear and cytoplasmic proteins can also be targets of specialized glycosyltransferases. While the significance of monosaccharide modifications of nucleocytoplasmic proteins is becoming appreciated in animals and plants, *Dictyostelium* is the first organism documented to express a complex nucleocytoplasmic O-glycosylation pathway. Consisting of 5 glycosyltransferase activities, the Skp1 modification pathway is dedicated to a single protein, a subunit of the E3(SCF) class of polyubiquitin ligases. The pentasaccharide is mounted on a hydroxyproline – the product of a prolyl 4-hydroxylase that is regulated by oxygen availability. This modification forms the molecular basis of an oxygen-sensing pathway that controls fruiting body formation. A second and recently discovered form of nucleocytoplasmic glycosylation consists of O-fucose on Ser and Thr residues. O-fucosylation is mediated by Spy, a glycosyltransferase that is highly homologous to the O-GlcNAc-transferase which catalyzes O-GlcNAcylation and mediates stress and other responses in animals. O-Fuc occurs on multiple nucleocytoplasmic proteins, based on Western blot analysis with *Aleuria aurantia* lectin, and immunofluorescence studies reveal a predominant nuclear localization. Disruption of *spy* has no major effects under laboratory conditions, and overexpression of Spy restores O-Fuc to above wild-type levels. We also discovered nucleocytoplasmic glycosylation in unrelated protozoans, including *Toxoplasma gondii*, the agent for human toxoplasmosis, and the crop pathogen *Pythium ultimum*. Skp1 glycosylation and O-fucosylation are each required for optimal *Toxoplasma* growth in a monolayer infection model, attesting to the functional significance of these conserved modifications across many protists.

## 9. Role of *PEPC* gene in prespore lineage choice and differentiation in *D. discoideum*

Kenichi Abe[1], Satoshi Kuwana[2], Kazuteru Taoka[1], Hidenori Hashimura[3], Satoshi Sawai[34], Masashi Fukuzawa[1]
[1]Faculty of Agriculture and Life Science, Hirosaki University, Japan
[2]Dept of Genetics, Evolution and Environment, University College London, UK
[3]Graduate School of Arts and Sciences, University of Tokyo, Japan
[4]Research Center for Complex Systems Biology, Universal Biology Institute, University of Tokyo, Japan

During development of multicellular organisms, it is still unclear how cell lineages are primed and their own cell fates are retained towards final differentiation. To address these questions, we used *D. discoideum* to know how undifferentiated cells choose and thereafter maintain their own cell fate(prestalk/prespore) during development. Among a number of REMI mutants, we identified a mutation in the gene encoding phosphoenolpyruvate carboxylase (*pepc*) that functions in glycolysis. The *pepc*- mutant cells aggregated normally, but later exhibited a typical “slugger” phenotype. In the *pepc*- mutant slugs, expressions of a prespore-specific marker gene (*pspA*p:RFP) and prespore-specific antigen (PSV) are abolished, suggesting that *pepc* contributes to prespore cell differentiation.

Expression of a reporter gene driven by *pepc* promoter (*pepc*:GFP) was observed in prestalk region and more strongly in prespore region in wild-type slugs. Interestingly, heterogeneous expression of *pepc*:GFP was observed in a clonal cell population during the vegetative stage.

To investigate the relation between heterogeneous expression of *pepc* in vegetative cells and prespore-biased expression in development, we performed the Halo-tag pulse chase assay by labeling vegetative cells highly expressing *pepc* to trace their lineage. Those labelled cells were distributed in whole of the slug, but showed spore-biased differentiation. These results suggest that *pepc* contributes to the heterogeneity of growing cells for cell lineage choice and has a role in cell type specification into prespore/spore cell during multicellular development.

**10. Cellular plasticity in *Dictyostelium***

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Dedifferentiation is a critical response to tissue damage. Insight into mechanisms underlying the reversal of differentiation will open substantial avenues for R&D in regenerative medicine. In contrast to most mammalian dedifferentiation systems, *Dictyostelium* cells show an unusually rapid and efficient dedifferentiation response. To understand how *Dictyostelium* dedifferentiate so well, we have carried out a combination of transcriptomic, imaging and molecular genetic studies to decipher the fundamental control features of cell plasticity. I will present the transcriptomic time series data cataloguing the multiple phases of dedifferentiation. Markers from these time series have been compared to cell biological transitions observed using live imaging, to align the dependencies between different milestones of dedifferentiation. Using single cell transcriptomics, we have determined features of dedifferentiation for stalk and spore lineage cells. Finally, I will present the initial results of a focussed screen of candidate regulators of cell plasticity.

### 11. Volatile Signals Synchronize Development in Social Amoebae

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Synchronicity is a necessary component of effective regulation and function in most biological organisms, and few organisms exhibit higher developmental synchrony than the eukaryotic soil amoeba, *Dictyostelium discoideum*. In this study, we disrupted the conserved putative transcription factor, *gtaI*, and characterized its role in regulating developmental synchrony. We found that mutating *gtaI* results in an increased sensitivity to volatile signaling molecules including ammonia and terpenoids. Low levels of ammonia were sufficient to severely impede development, while altered terpene composition in the mutant headspace resulted in late-stage developmental delay and asynchrony. We further determined that the accumulation of at least one terpene was compromised by the mutation and demonstrated that synchronicity can be restored by introducing wild-type volatile bouquet into the mutant headspace. In so doing, we have discovered a novel regulatory process by which developmental synchrony is achieved in social amoeba via volatile signaling.

## 12. Eco-evolutionary significance of 'loners'

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When a population of *Dictyostelium discoideum* cells starves, most of these cells aggregate and eventually develop into the fruiting body. A portion of these cells, however, never joins the aggregate, staying behind as loners. If food is replenished to the environment, loner cells promptly grow, divide, and recapitulate development. Here we show that natural populations of *D. discoideum* have heritable variation in their loner allocation, and that such allocation depends on biotic and abiotic conditions. We build a simplified model of aggregation to show how loner patterns might emerge and point towards the pre-aggregation stage of development as the critical point where loners are determined. Finally, we embed this developmental model into an ecological model spanning multiple starvation cycles and show that the developmental interactions between strains regarding loner allocation might have conflicting effects for biodiversity maintenance across spatial and temporal scales.

## 13. Recent insights into Cln5 and Mfsd8 function in *Dictyostelium*

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The neuronal ceroid lipofuscinoses (NCLs), collectively known as Batten disease, are progressive neurological disorders characterized by the buildup of intracellular material within neurons. The disease, which affects all ages and ethnicities, is classified into 13 different subtypes based on the specific gene that is mutated (*CLN1-8*, *CLN10-CLN14*). The *Dictyostelium* genome encodes homologs of 11 of the 13 NCL proteins and has emerged as a powerful model system for studying the localization and functions of these proteins. Our current work is focused on understanding the cellular role of Cln5 and Mfsd8 in *Dictyostelium*, which are homologs of human CLN5 and MFSD8/CLN7, respectively. We previously showed that *Dictyostelium* Cln5 is secreted and has glycoside hydrolase activity. We are using this information to study the role of the protein in regulating autophagy and multicellular development. Our recent work on Mfsd8 suggests that the protein plays an important role in regulating growth and the early stages of development. The goal of this work is to reveal phenotypes in *Dictyostelium* that can be translated to mouse and human cell models to ultimately develop effective treatments for this devastating form of neurodegeneration.

## 14. GGPP Depletion Selectively Kills Cells Lacking PTEN by Blocking Motility and Macropinocytosis

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PTEN is one of the most frequently mutated tumor suppressors in human cancer. Loss of PTEN leads to hyperactivation of the PI3K–AKT signaling pathway, cell migration, and cell proliferation. Aiming to identify specific drugs that could target cancer cells, we screened 2650 FDA approved compounds and discovered that pitavastatin selectively kills *Dictyostelium pten-* cells. This result carries over to mammalian cells in both 2D and 3D culture: MCF10A cells lacking *PTEN* are more sensitive to pitavastatin. Mevalonic acid and geranylgeranyl pyrophosphate (GGPP), but not farnesyl pyrophosphate or squalene, rescued pitavastatin treated MCF10A *pten-/-* cells. Consistently, *Dictyostelium ggps1*$^-$ cells showed defects in both cell migration and fluid uptake, suggesting that GGPP is required for the formation of pseudopods and macropinosomes. Furthermore, compared to wild type cells (AX2 and MCF10A), the macropinocytosis in *PTEN* deleted cells showed more defects and higher sensitivity to GGPP depletion. Macropinocytosis of proteins serves as an essential mechanism for nutrient uptake, which may explain why *PTEN* deleted cells are more sensitive to pitavastatin. Finally, we found pitavastatin significantly decreased Twist1-induced cell dissemination and killed breast cancer organoids in 3D mouse model. These findings implicate pitavastatin as a potential therapeutic candidate for tumors.

## 15. Autophagy is upregulated in a *Dictyostelium* model of CLN5 disease

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The neuronal ceroid lipofuscinoses (NCLs) are a group of neurodegenerative, lysosomal storage disorders known as Batten disease. There are 13 genetically distinct subtypes of NCL, each one being caused by a mutation in a specific gene (*CLN1* to *CLN8*, *CLN10* to *CLN14*). The pathological mechanisms underlying Batten disease are unknown because the functions of the NCL proteins are not well understood. Previously, we showed that a homolog of one of these proteins in *Dictyostelium*, Cln5, is secreted through an unconventional mechanism linked to autophagy. Based on these findings, we sought to further investigate the role of Cln5 in autophagy by using a *Dictyostelium cln5* knockout model (*cln5*$^-$). We observed that loss of *cln5* reduces cell proliferation in nutrient-limiting media during growth, causes cells to develop precociously, and results in aberrant development on agar containing the autophagy inhibitor, ammonium chloride. These observations led us to investigate the intracellular mechanisms regulating autophagy in *cln5*$^-$ cells. We observed that *cln5*-deficiency increases the amount of the autophagosome marker Atg8b in starved cells. In an RFP-GFP-Atg8 autophagic flux assay, we found that *cln5*-deficiency increases the number of autophagosomes and lysosomes. Together, these results indicate that loss of *cln5* induces autophagy and suggests that Cln5 may play an important role in regulating autophagy in *Dictyostelium*. The cellular processes that regulate autophagy in *Dictyosteli*um are similar to those that regulate the process in mammalian cells. Thus, this research provides insight into the undefined pathological mechanism of CLN5 disease and could identify cellular pathways for targeted therapeutics.

## 16. *Dictyostelium discoideum* as a model for Parkinson's disease?

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Current treatments for Parkinson's disease (PD) only alleviate symptoms doing little to inhibit the onset and progression of the disease, thus we must learn more about the underlying mechanism of this disease. Rotenone is a known inducer of parkinsonian conditions in rats and we hypothesized it could induce parkinsonian cellular conditions in *Dictyostelium discoideum.* We use *D. discoideum* to primarily focus on mitochondrial dynamics which are linked to many neurological diseases, thus we also analyzed dynamics in the presence of rotenone. Our study shows that rotenone disrupts the actin and microtubule cytoskeleton but mitochondrial morphology remains intact. Rotenone stimulates mitochondrial velocity while inhibiting mitochondrial fusion, increases reactive oxygen species (ROS) but has no effect on ATP levels. Antioxidants have been shown to decrease some PD symptoms thus we added ascorbic acid to our rotenone treated cells. Ascorbic acid administration suggests that rotenone effects may be specific to the disruption of the cytoskeleton rather than the increase in ROS in *D. discoideum*. Our results imply that *D. discoideum* may be a valid cellular PD model and that the rotenone induced velocity increase and loss of fusion could prevent mitochondria from effectively providing energy and other mitochondrial products in high demand areas. The combination of these defects in mitochondrial dynamics and increased ROS could result in degeneration of neurons in PD.

**17. Polyphosphate inhibits the killing of ingested bacteria in *Dictyostelium* and macrophages**

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Polyphosphate is a linear chain of phosphate residues and is present in organisms ranging from bacteria to humans. Many pathogenic microbes accumulate polyphosphate, and mutation of the gene encoding the polyphosphate kinase that synthesizes polyphosphate decreases their polyphosphate production and decreases their survival. How polyphosphate potentiates pathogenicity is poorly understood. We found that *Dictyostelium* cells accumulate extracellular polyphosphate at high cell densities, and this inhibits their proliferation possibly by inhibiting digestion of internalized nutrients. *E. coli* bacteria accumulate undetectable levels of extracellular polyphosphate, and have poor survival after phagocytosis by *Dictyostelium* or human macrophages. In contrast, *Mycobacterium smegmatis* bacteria accumulate detectable levels of extracellular polyphosphate, and have relatively better survival after phagocytosis by *Dictyostelium* or macrophages. Adding extracellular polyphosphate increased *E. coli* survival after phagocytosis by *Dictyostelium* and macrophages. Reducing expression of polyphosphate kinase 1 in *M. smegmatis* reduced accumulation of extracellular polyphosphate and reduced survival in *Dictyostelium* or macrophages, and this was reversed by addition of extracellular polyphosphate. Conversely, treatment of *Dictyostelium* and macrophages with recombinant yeast exopolyphosphatase reduced survival of *M. smegmatis*. *Dictyostelium* cells lacking the putative polyphosphate receptor GrlD had reduced sensitivity to polyphosphate, and compared to wild-type cells showed increased killing of phagocytosed *M. smegmatis*. Polyphosphate inhibited phagosome acidification and maturation in *Dictyostelium* cells, and reduced early phagosomal markers, early endosomal antigen 1, and Rab 5 protein in macrophages infected with mycobacteria. Our findings identify polyphosphate as a possible signal that some bacteria use to inhibit phago-lysosomal killing to survive in host cells.

## 18. Discoidin I regulation of bacterial carriage during the growth to development transition in *Dictyostelium discoideum*.

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*Burkholderia* bacteria infectiously induce the proto-farming behavior of wild isolates of *Dictyostelium discoideum* that allows them to carry food bacteria through development (1,2). Non-carrier amoebae initiate when the bacteria are depleted and complete development axenically, whereas the amoebae of carrier strains begin development before they exhaust the bacterial food supply. We have found that carriers secrete carbohydrate-binding proteins (lectins) during the growth to development transition that influence amoebal survival and induce bacterial carriage and endosymbiosis (3,4). The discoidin I lectins Discoidin A (DscA) and Discoidin C (DscC) bind to and protect bacteria from killing by antibacterial proteins secreted by *D. discoideum* and are also sufficient to induce bacterial endosymbiosis, resulting in live bacteria within amoebae and spores (3). CadA is another lectin secreted by carrier strains that mediates bacterial agglutination and appears to be required for the formation of a transient physical barrier between growing amoebae and bacteria (4).

Given their role in bacterial carriage amoebal survival we have investigated the mechanism of DscA/C and CadA production during the growth to development transition. We will show that DscA appears to act as an extracellular signal that induces additional discoidin production by amoebae. This leads to the testable hypothesis that bacterial carriage results from a change in the sensitivity of the discoidin signaling pathway. Since discoidin binds to both bacteria and amoebae, increased sensitivity of amoebae to discoidin should lead to discoidin production when the bacteria to amoebae ratio is high, as is observed in carrier strains. Our work suggests that autocrine regulation of discoidin I production is a central feature of the growth to development transition in *D. discoideum*.

1. Brock, D.A., T. E. Douglas, D. C. Queller and J. E. Strassmann (2011). Primitive agriculture in a social amoeba. *Nature* 469:393-396.
2. DiSalvo S., Haselkorn T.S., Bashir U., Jimenez D., Brock D.A., Queller D.C., Strassmann J.E. (2015) Burkholderia bacteria infectiously induce the proto-farming symbiosis of Dictyostelium amoebae and food bacteria. *Proc Natl Acad Sci U S A* 112:E5029-5037.
3. Dinh, C., Farinholt, T., Hirose, S., Zhuchenko, O. and A. Kuspa. 2018. Lectins modulate the microbiota of social amoebae. *Science* 361:402-406.
4. Farinholt, T., Dinh, C. and A. Kuspa. 2019. Social amoebae establish a protective interface with their bacterial associates by lectin agglutination. *Science Advances*, in press.

## 19. How amoebae locate and eat bacteria-- G-protein-coupled receptor mediated cell migration and bacterial engulfment

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Phagocytic cells locate microorganisms via chemotaxis, then consume them using phagocytosis. Cells of the social amoeba Dictyostelium are stereotypical phagocytes that prey on diverse bacteria using both processes. However, as the typical phagocytic receptors, such as complement receptors and Fc gamma receptors used by mammalian phagocytes, have not been found in Dictyostelium, it remains mysterious how these cells recognize bacteria. We developed a quantitative phosphoproteomic technique to uncover signaling components. Using this approach, we discovered an orphan class C G-protein-coupled receptor (GPCR) as the long sought-after folate receptor, fAR1, in Dictyostelium. Specifically, fAR1 simultaneously recognizes the diffusible chemoattractant folate and the phagocytic cue lipopolysaccharide (LPS), a major component of the bacterial coat. We found that cells lacking the receptor fAR1 or its cognate G-proteins are defective in chemotaxis toward folate and phagocytosis of Klebsiella aerogenes. Recently, we have purified the extracellular domain of fAR1 and solved its crystal structure. It folds as a Venus-Flytrap (VFT) domain, acting as the binding site for both folate and LPS. Thus, fAR1 represents a new member of the pattern recognition receptors (PRRs) and mediates signaling from both diffusible chemoattractants and bacterial surface molecules to reorganize actin for chemotaxis and phagocytosis. Currently, we are in the process of characterizing another GPCR as the potential receptor for gram positive bacteria recognition. Therefore, GPCRs could be the most ancient receptor family detecting microbial associated molecular pattern during host pathogen interaction.

**20. AlyL: a *D. discoideum* putative lysozyme with an antibacterial activity Tania**

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Lysozymes are enzymes which hydrolyse the bond between N-acetylglucosamine and N-acetylmuramic acid. Such bonds are found in a major component of the bacterial cell wall called the peptidoglycan. Lysozymes have also been described as having a bacteriolytic activity against Gram-positive and Gram-negative bacteria. Surprisingly it has been convincingly demonstrated that lysozyme's enzymatic and bacteriolytic activities are distinct. *D. discoideum* is equipped with a large arsenal of antimicrobial peptides, including 22 putative lysozymes belonging to four different families: amoeba lysozymes (Aly), *Entamoaba histolytica* lysozymes, T4-phage lysozymes, and chicken lysozymes. This study focuses on the amoeba lysozymes. In *D. discoideum*, there are eight Aly proteins: AlyA, AlyB, AlyC, AlyD-1, AlyD- 2, AlyL, AlyTM1, and AlyTM2. AlyA has been shown to be responsible for most of *D. discoideum*'s lysozyme activity.

By generating *alyA*-knockout (KO) and *alyL*-KO mutants, we show that AlyL is necessary for efficient intracellular killing of *Klebsiella pneumoniae* bacteria, but not a major contributor to the lysozyme activity of *D. discoideum*. Symmetrically, we show that AlyA represents a large fraction of *D. discoideum* lysozyme activity but is not necessary for intracellular killing of *K. pneumoniae.* We are currently delineating the segment of AlyL that is implicated in its bacteriolytic activity.

**21. Evolutionary battles for iron between Dicty and Legionella**

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During infections, intracellular pathogens must acquire all of their nutrients from within host cells. Therefore, one common defense strategy for eukaryotic hosts is to sequester essential nutrients like iron, thereby starving pathogens and restricting their growth. This dynamic can set up an evolutionary arms race between host and microbe, a molecular tug-or-war competition for nutrients that ultimately shapes the mechanisms of iron acquisition in both organisms. Integrating evolutionary and functional studies, we investigate how such battles for iron have played out in Dictyostelium and in Legionella, specifically focusing on adaptation within the Legionella iron transporter MavN.

**22. Dual RNA-seq of naïve *D. discoideum* and their *Burkholderia* endosymbionts hints at different evolutionary histories**

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*Dictyostelium discoideum* is an established model for host-pathogen interactions as well as an emerging model for host-symbiont relationships. Roughly a quarter of *D. discoideum* isolated from the wild carry a *Burkholderia* bacterial symbiont. These facultative endosymbionts cause infected amoebae to carry food bacteria across their starvation-induced multicellular cycles. We model the initial establishment of this symbiosis by comparing gene expression patterns of naive hosts and *Burkholderia* endosymbionts isolated from amoebae, in the presence and absence of the other party. We tested the effect of the two clades of *D. discoideum*-associated *Burkholderia*, *Ba*-clade vs. *Bh/Bb*-clade, combined with three wild amoebae clones. Amoebae infected with *Burkholderia* up-regulate genes related to endocytosis, various metabolic processes, and defense responses against bacteria in common. But amoebae infected with *Bb* change expression of only a small proportion of their genome (13%) while amoebae infected with *Bh* and *Ba*-clade show more changes (30 - 34%). *Burkholderia* change the expression of a large portion of their genomes (46 - 59%) when inside *D. discoideum* compared to when they are free-living. Up-regulated classes of genes include carbohydrate transport systems in *Ba*-clade and protein metabolic processes in *Bh/Bb*-clade *Burkholderia*. We hypothesize that the differences in both host and symbiont in their responses, as well as differences in genome sizes between the clades, may be due to a longer evolutionary association between social amoebae and *Bh/Bb*-clade compared to *Ba*-clade *Burkholderia*.

**23. Fine tuning of the Anionic Membrane Lipids PI(4,5)P2 and Phosphatidylserine Establishes Polarized Morphologies and Regulates Cell Migration**

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The ability of the cell to break symmetry and properly establish the polarized localization of signaling molecules is critical for division, migration, axon guidance and various immune responses. *Dictyostelium* has been a powerful model system for understanding how phosphoinositide (PI)-mediated signaling pathways set up these polarity circuits. When the enzymes regulating these PIs are spatially or temporally misregulated, pathologies can occur, including tumorigenesis. We previously found that PI(4,5)P2 levels set a threshold for cell excitability. PI(4,5)P2 levels below a threshold trigger Ras GTPase and PI3K activity, contributing to signaling networks that lead to cell protrusions. Above-threshold PI(4,5)P2 levels support regulators that contribute to quiescent membrane activity and cell retraction. In addition, we show a correlation between the PI(4,5)P2 threshold and rates of phosphatidylserine (PS) exposure on the outer leaflet of the plasma membrane (PM). Receptor-mediated cell stimulation could trigger this PS exposure. Interestingly, the same responses were achieved upon the synthetic lowering of the PM PI(4,5)P2 levels using a chemical dimerization system. Taken together, this work demonstrates a mechanistic link between the PM PI(4,5)P2 levels and the rate of PS exposure on the periphery of the cell. The changes in the two together likely play a major role in the electrostatic interactions on the inner leaflet of the plasma membrane and regulate the the binding and signaling of many proteins at the PM. These findings in *Dictyostelium* show the importance of using this model system to further our understanding of cell polarity. This work also suggests that the phosphoinositide PI(4,5)P2 and the phospholipid PS should be considered novel therapeutic targets for fighting tumorigenesis and cancer metastasis.

## 24. GoldenBraid cloning for synthetic biological applications in *Dictyostelium*

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Synthetical biological applications in *Dictyostelium* often use common elements such as antibiotic resistance cassettes, constitutive or conditional promoters, and fluorescent proteins. Cloning these elements in the proper order and context can be a laborious, multistep process that has to be repeated for every new construct. GoldenBraid is a cloning system that combines the robustness and speed of the GoldenGate cloning reaction with a standardized grammar to modularly assemble genetic parts into transcriptional units, and then combine multiple transcriptional units into single plasmids(1). GoldenGate and GoldenBraid cloning reactions are one-test-tube, combined restriction digestions and ligations that enrich for a desired cloning product. The GoldenBraid grammar takes the form of a standardized set of 4-base-pair sticky ends that flank genetic parts and assemblies of parts. A given part is initially domesticated (i.e. made ready for use within the GoldenBraid system) as a single insert in a standard vector backbone. A given part's flanking sticky ends are generated by digestion with a Type IIS restriction enzyme, which cuts DNA outside its recognition site. In GoldenBraid vectors, restriction sites for two Type IIS enzymes, BsmBI and BsaI, flank inserts in an overlapping (i.e. braided) orientation. The particular sequences of the 4-base-pair sticky ends define the identities of parts (promoters, open reading frames, protein tags, terminators, etc.) such that they can be modularly assembled into transcriptional units in a dedicated set of backbones. These units, in turn, can be iteratively combined (up to 5 in a single GoldenBraid reaction). To minimize assembly steps, we have modified the system's assembly backbones so that each contains one of three *Dictyostelium* antibiotic resistance cassettes (NeoR, HygR, and BsR). We have also domesticated a library of useful genetic parts and validated them in assemblies of transcriptional units. These include 6 promoters, 8 fluorescent proteins, 5 epitope tags, 2 linkers, 4 fluorescent-protein-based sensors, 2 luciferases, 40 barcodes, 3 terminators, Cas9, and HypaCas9. We domesticated the fluorescent proteins as N-terminal tags, stand-alone ORFs and C-terminal tags, and the epitope tags as both N-terminal and C-terminal tags. We intend to deposit this library of parts and assemblies in the Dicty Stock Center in the hopes that most researchers in the field will start using the system as a platform for rapid and efficient cloning with standard parts. Adding new components to the system involves PCR amplification of the desired element to add the grammatical overhangs and remove Type IIS restriction sites within the part, followed by GoldenBraid cloning into the standard domestication vector backbone. Subsequent assembly also employs this one-test-tube cloning format. We propose that *Dictyostelium* researchers will deposit their new parts and assemblies in the Stock Center, which will allow all of us to generate new vector combinations in a standardized manner. During the workshop, we will review the principles of the system and solicit input from the participants.

1 Sarrion-Perdigones A, Palaci J, Granell A, Orzaez D. 2014. Design and construction of multigenic constructs for plant biotechnology using the GoldenBraid cloning strategy. Methods in molecular biology (Clifton, NJ). 1116:133-51.

## 25. dictyBase Version 2 – work in progress

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The new dictyBase is now slowly emerging. We will represent the new login workflow which currently allows our curators to directly edit the webpage content of dictyBase. This editing workflow will be extended for dictyBase users to perform community curation in the near future. The new database also allows us to represent new data. For example, we already imported and display more expressive Gene Ontology annotations that curators have been annotating offline in the past couple of years. Annotated are now relationships between GO terms, and between gene products and GO terms. For example, a gene is *involved_in* **activation of GTPase activity** and *has_regulation_target* (rapA) and *occurs_at* **cell leading edge cortex**

Our next major release focusses on the Dicty Stock Center: strains including the new GWDI mutants, phenotypes of the strains, cited literature, and the ordering process via shopping cart with newly generated forms and ordering confirmations. This will make ordering and order processing more efficient. The voluntary user login here will further streamline DSC orders for users.

We will demonstrate our newest annotations and tools and encourage the user community to ask questions and / or provide input.

Acknowledgements: This work is supported by NIH/NIGMS grants GM064426 and GM087371

## 26. Phosphorylated Rho-GDP Directly Activates mTORC2 Kinase Toward AKT Through Dimerization with Ras-GTP to Regulate Cell Migration

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mTORC2 plays critical roles in metabolism, cell survival, and actin cytoskeletal dynamics via phosphorylation of AKT. Despite its importance to biology and medicine, it is unclear how mTORC2-mediated AKT phosphorylation is controlled. Here, we identify an unforeseen principle by which a GDP-bound form of the conserved small G protein Rho GTPase directly activates mTORC2 in AKT phosphorylation in social amoebae *Dictyostelium* cells. Using biochemical reconstitution with purified proteins, we demonstrate that Rho-GDP promotes AKT phosphorylation by assembling the supercomplex with Ras-GTP and mTORC2. This supercomplex formation is controlled by chemoattractant-induced phosphorylation of Rho-GDP at serine 192 by GSK-3. Furthermore, Rho-GDP rescued defects in both mTORC2-mediated AKT phosphorylation and directed cell migration in Rho-null cells in a manner dependent on phosphorylation of serine 192. Thus, in contrast to the prevailing view that GDP-bound forms of G proteins are inactive, our study reveals that mTORC2-AKT signaling is activated by Rho-GDP.

## 27. Adhesion strategies of Dictyostelium discoideum – a force spectroscopy study

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Biological adhesion is essential for all motile cells and generally limits locomotion to suitably functionalized substrates displaying a compatible surface chemistry. However, organisms that face vastly varying environmental challenges require a different strategy. The model organism Dictyostelium discoideum (D.d.), a slime mould dwelling in the soil, faces the challenge of overcoming variable chemistry by employing the fundamental forces of colloid science. To understand the origin of D.d. adhesion, we realized and modified a variety of conditions for the amoeba comprising the absence and presence of the specific adhesion protein Substrate Adhesion A (sadA), glycolytic degradation, ionic strength, surface hydrophobicity and strength of van der Waals interactions by generating tailored model substrates. By employing AFM-based single cell force spectroscopy we could show that experimental force curves upon retraction exhibit two regimes. The first part up to the critical adhesion force can be described in terms of a continuum model, while the second regime of the curve beyond the critical adhesion force is governed by stochastic unbinding of individual binding partners and bond clusters. We found that D.d. relies on adhesive interactions based on EDL-DLVO (Electrical Double Layer-Derjaguin–Landau–Verwey–Overbeek) forces and contributions from the glycocalix and specialized adhesion molecules like sadA. This versatile mechanism allows the cells to adhere to a large variety of natural surfaces under various conditions.

## 28. The Atypical MAP Kinase ErkB Transmits Distinct Chemotactic Signals through a Core Signaling Module

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Signaling from chemoattractant receptors activates the cytoskeleton of crawling cells for chemotaxis. We show using phospho-proteomics that different chemoattractants cause phosphorylation of the same core set of around 80 proteins in *Dictyostelium* cells. Strikingly, the majority of these are phosphorylated at an [S/T]PR motif by the atypical MAP kinase ErkB. Unlike most chemotactic responses, ErkB phosphorylations are persistent and do not adapt to sustained stimulation with chemoattractant. ErkB integrates dynamic autophosphorylation with chemotactic signaling through G-protein-coupled receptors. Downstream, our phosphoproteomics data define a broad panel of regulators of chemotaxis. Surprisingly, targets are almost exclusively other signaling proteins, rather than cytoskeletal components, revealing ErkB as a regulator of regulators rather than acting directly on the motility machinery.

**29. Negative Surface Charge Defines the State of Cell Cortex and Regulates Excitable Dynamics**

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The spatially distributed components of signal transduction excitable network (STEN) and cytoskeletal excitable network (CEN) in a Dictyostelium cell generate autonomous waves in the substrate attached basal surface. Similar autonomous waves underlie crucial physiological processes such as phagocytosis, cytokinesis, macropinocytosis, and pseudopodia-based cell motility. One key event in this characteristic cortical wave formation is the symmetry breaking of membrane phospholipids and self-organization of STEN and CEN proteins to dynamically generate distinct regions in the cell cortex which corresponds to the typical front (or “activated”) and back (or “inactivated”) regions of a randomly migrating cell. An analogous pattern of organization is observed in chemotactic migration, micropinocytosis, and cytokinesis of different types of eukaryotes cells. We have identified two novel unexpected features of these self- organizing patterns. First, we have found that three lipidated membrane proteins, PKBR1, Gßγ, and RasG, which are widely thought to be uniformly distributed over the membrane, show preferential localization in the back/inactivated regions of cell cortex. Our data indicates that besides conventional recruitment based mechanism used by many other components, a dynamic partitioning process may contribute to the self- organization pattern of components that remain associated with the membrane. Second, we have assessed the localization of the generic surface charge sensor that was previously shown to be tightly associated with the negatively charged inner surface. We found that the sensor partitioned to the back/inactivated region as did the lipidated proteins. We propose that high negative surface charge defines the back or inactivated region of the cell cortex and thus cell migration and macropinocytosis is essentially guided by the alteration of surface potential on the inner leaflet to generate dynamic polarity.

## 30. The torsinA homologue tsin is required for the multicellular development of *Dictyostelium discoideum*

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DYT1 dystonia is a neurological movement disorder characterized by repetitive muscle contractions that result in involuntary twisting of the extremities and abnormal posturing. DYT1 dystonia is caused by a mutation within the DYT1/Tor1a gene that deletes a single glutamic acid residue (ΔE302/303 or ΔE) from the evolutionarily conserved AAA+ protein torsinA that localizes to the shared lumen of the endoplasmic reticulum and nuclear envelope. The mechanism through which the ΔE mutation causes DYT1 dystonia is unclear because the basic cellular function of torsinA is unknown. A significant hindrance in our understanding of torsinA function has been the lack of torsinA homologues in well-established, experimentally tractable single-cell model systems like yeast. Here we report the identification of a torsinA homologue, TorSIN (tsin), in the social amoeba *Dictyostelium discoideum* (Dicty). GFP-tagged wild type tsin localizes to the endoplasmic reticulum and nuclear envelope. Like torsinA, mutation of the predicted Walker B motif that is responsible for ATP-hydrolysis in AAA+ proteins results in accumulation of GFP-tsin within the nuclear envelope indicating that potential tsin substrates reside within this subcellular compartment. Under favorable environmental conditions, dicty exist as free-living amoeba that grow and divide by mitosis. Upon starvation, individual amoeba will aggregate by polarizing and migrating towards a secreted cAMP chemoattractant resulting in the formation of a small mound that differentiates to form a spore-containing fruiting body. To test the potential function of tsin during dicty migration and development, we used homologous recombination to disrupt tsin expression. Tsin-null cells exhibit striking defects in during early aggregation, with significantly smaller aggregation territories than wild type controls. Additionally, while tsin mutant cells can aggregate and eventually form mounds, they have a defect in the transition out of the mound phase, with only a fraction of mounds developing into fruiting bodies. Surprisingly, tsin-null cells migrate faster than wild type cells and are competent for chemotaxis when cAMP is experimentally provided. Taken together, these results suggest that tsin is required for signaling propagation during multicellular development in Dicty. Furthermore, they establish Dicty as a powerful model system for the dissection of the evolutionarily conserved functions of torsinA.

**31. Proteomic analysis of chemoattractant-induced signaling dynamics**

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Cell migration plays an important role in various biological and pathological conditions. Migrating cells often adopt a polarized state with morphologically and functionally distinct leading and trailing edges. This process relies on the selective activation of key signaling molecules on the plasma membrane. Intriguingly, some of these molecules are encoded by oncogenes and tumor suppressors, respectively, and display characteristic behaviors in cells stimulated with chemoattractants or undergo morphological changes in general, such as macropinocytosis, phagocytosis, and cell division. Utilizing this unique feature, we designed proteomic experiments to isolate peripheral membrane proteins that function as potential signaling molecules. A series of new factors that are spatially restricted to either end of the cell were identified, and among which, a protein we named Crtp2. Crtp2 localizes at the cell's leading edge, including structures of protrusions and macropinosomes, by binding to PI(3,4)$P_2$ and PIP3. Its overexpression promotes abnormal filopodia formation, resulting in impaired cell migration and macropinocytosis. The null cells are also defective in growth and macropinocytosis. Using co-immunoprecipitation experiment, we discovered that one of the interactors of Crtp2 is the actin nuclear promoting factor, the SCAR/WAVE complex. We propose that Crtp2 may function as a linker between the cell membrane and the actin cytoskeleton network to regulate leading edge activities.

## 32. Roles of typical and atypical MAP kinases in *Dictyostelium*

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MAP kinases (MAPKs) play important roles in many eukaryotic signaling pathways including responses to chemoattractants. *Dictyostelium* encodes two MAPKs, Erk1 and Erk2, that can be classified as typical and atypical MAPKs, respectively, based on their sequence and regulation. Chemoattractants, folic acid and cAMP activate Erk2 in an early response and then Erk1 in a secondary response that requires Erk2. Loss of Erk2 abolishes chemotaxis to both chemoattractants whereas loss of Erk1 only slightly impairs chemotaxis. Loss of both MAPKs reduces cell motility and impairs growth. The only conventional MAPK kinase in *Dictyostelium*, is only required for the activation of Erk1 and not Erk2, supporting the classification of Erk2 as an atypical MAPK. The activation and interactions of atypical MAPKs remains to be fully characterized in any organism but we have recently found that the GtaC transcription factor is an *in vitro* substrate of Erk2. GFP-tagged GtaC can be used as a real-time reporter for Erk2 function because the shuttling of GFP-GtaC from the nucleus to the cytoplasm in response chemoattractants is dependent on Erk2 function.

## 33. Identifying the knobs used to modulate collective signaling in *Dictyostelium*

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Coordinating biological behaviors across groups of cells is critical for a wide range of biological processes ranging from development to wound healing. How these basic group phenomena are regulated, potentially by modulating factors like the frequency of synchronized signaling or the speed of group migration at the level of single cells is still an open question. Identifying what single cells tune in their own signaling programs to produce these phenotypic changes in group-wide behaviors would yield parameters we can control when reprogramming these systems for our benefit. To address this challenge, we are pursuing two complimentary efforts in the classic model system for collective signaling, *Dictyostelium discoideum*. First, we are interrogating cAMP signaling behaviors in mutant strains to link specific signaling network components to single-cell dynamics that are known to lead to changes in group-wide behaviors. Second, we are linking single-cell signaling to population-wide behaviors through direct experimental optogenetic control over cAMP signaling. Through directly controlling signaling, we can causally link our observations of single-cell signaling dynamics to the population-wide behaviors they control. Together, these efforts will allow us to identify how population-wide multicellular behaviors are regulated at the level of single-cell signaling.

**34. Reduced oxygen availability triggers aerotactic migration in Dictyostelium**

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A state of low oxygen occurs frequently in soil, water and multicellular tissues and has played a pivotal evolutionary role in shaping multicellularity. While Dictyostelium is an obligatory aerobic organism, its ecological niche in the soil and around large amount of bacteria will expose the amoeba to reduced oxygen availability. We have recently observed that vertically confining a micro-colony of *Dictyostelium* cells in a growth medium triggers cells to move quickly outward of the self-generated central hypoxia area and thus form an expending ring. The analysis of the cells behavior within the micro-colony reveals a complex response to hypoxia depending upon their position:
- Cells at the very hypoxic center are immobile but remains viable even after three days.
- Cells closer to the ring present a clear outward directionality.
- Cells at intermediate positions are very polarized and motile but with limited outward directionality
- Cells within the outer part of the ring are poorly polarized as typical vegetative cells

Using a fluorescent oxygen sensor included in a thin PDMS film, we can show that the ring of cells occurs at the level of a sharp oxygen gradient. The various cell behaviors in the micro-colony was mimicked using an in silico cellular POTTS model that includes both a positive aerotaxie upon an oxygen gradient but also a differential response to the absolute oxygen concentration. The signaling pathway behind aerotaxie remains unclear and is still under investigation.

## 35. TalA, MhcA, and SCAR regulate blebs in response to environmental compression in *Dictyostelium discoideum*

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Blebbing, the dominant form of cell movement in 3D environments, has become increasingly important in cellular and cancer biology. Unlike filopodia, lamellipodia, and pseudopodia that use actin-rich structures to push the membrane forward for movement, blebs use hydrostatic pressure to detach the membrane from the cortex for movement. Little is known about the proteins that regulate different motility structure use; however, we have identified three potential players. Talin, a cortex-to-membrane binding protein; Myosin II, an F-actin contracting motor protein; and SCAR/WAVE, a WASP family actin-nucleating protein have been implicated in bleb-based motility, but their impact on motility structure generation with respect to varying compression has not been thoroughly investigated. Using *D. discoideum* null lines (TalA, MhcA, and SCAR, respectively), we varied the pressure exerted on cells using two cAMP under agarose assays (0.4% and 0.7%) and quantified the cells' motility structure choice and the total number of structures. Previously, we found that wild-type cells increase bleb use at the expense of actin-based motility structures under higher pressure and that TalA is needed for cells to bleb efficiently under any pressure. Here, we show that MhcA is needed for bleb-based motility and that MhcA negatively regulates pseudopodia and lamellipodia under pressure. Lastly, SCAR positively regulates blebs and negatively regulates pseudopodia under lower pressure, but is not needed for bleb formation under high pressure. These results are being used to develop a mathematical model of bleb-based motility, which will help shed light on how the activities of these proteins influence bleb-based motility.

**36. How Does a Ras Inhibitor Control the Gradient-Sensing Dynamics and Chemotaxis in *Dictyostelium discoideum?***

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Dictyostelium discoideum cells chemotax through a large range of chemoattractant concentration gradients through a cellular process called adaptation. An adaptative cell no longer responds to the present stimulus but remains sensitive to higher concentration stimuli. Thus, it is believed that adaptation provides a fundamental mechanism for a long-range chemotaxis of eukaryotic cells. Recently, we identified Ras inhibitor C2GAP1 as a key inhibitor of Ras signaling required for chemoattractant-induced Ras adaptation and long-range chemotaxis in *Dictyostelium discoideum*. Here, we further show that C2GAP1 controls gradient-sensing dynamics in a concentration-dependent fashion. More importantly, C2GAP1 lowers the sensitivity of cells to chemotax through a higher concentration-range of chemoattractant gradient.

**37. Exosomes as key regulators of signal relay during chemotaxis**

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The property of sensing and initiating directional migration in response to external cues or chemotaxis is a fundamental property of biological systems. How cells detect and respond to external chemotactic signals and, in particular, how the spatial and temporal relay of chemotactic signals between cells impact single and group cell migration are key questions in the chemotaxis field. Using the social amoebae *Dictyostelium discoideum*, where cAMP acts as a chemoattractant, we have shown that the relay of chemotacitc signals between cells is mediated through the release of extracellular vesicles that contain the enzyme responsible for synthesizing cAMP, the adenylyl cyclase ACA. We purified the extracellular vesicles from chemotactic cells and showed that they are exosomal, contain and release cAMP and attract cells in an ACA-dependent fashion. Indeed, mass spectrometry analyses identified many canonical exosomal proteins as well as upstream regulators of ACA. We further show that cAMP is released through specific ABC transporters expressed in exosomes. We extended our studies to neutrophils and demonstrated that leukotriene $B_4$ ($LTB_4$), a key secondary chemoattractant in these cells, is also essential in mediating signal relay during chemotaxis both *in vitro* and *in vivo*. Here again, we show that the presence of $LTB_4$, as well as its synthesizing enzymes, in exosomes is essential to relay the chemotactic signals between neighboring cells as inhibition of exosome release leads to loss of directional motility with a concomitant loss of $LTB_4$ release. We envision that the packaging of chemoattractants in exosomes provides a means of maintaining highly diffusible signals available for long-range cell-cell communication. We foresee that this newly uncovered mechanism is used by other signals to foster communication between cells in harsh extracellular environments.

## Posters

### 1. Role of actin-binding proteins in sensing mechanical stimulation

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Molecular mechanisms by which cells sense and directionally migrate in response to mechanical perturbation are not well understood. *Dictyostelium discoideum* cells exposed to a mechanical stimulus show rapid and transient activation of multiple components of the signal transduction network, a response that requires an intact actin cytoskeleton of the cell. However, exactly what aspect of the actin cytoskeleton network is responsible for sensing and/or transmitting the signal is unclear. In this study, we investigated the role of the actin-binding protein filamin by analyzing shear flow-stimulated responses in filamin-null cells expressing full-length filamin, the actin-binding domain of filamin, or empty vector. Fluorescently-tagged Ras binding domain biosensor that detects active Ras was used as a readout of signal transduction network activation following exposure of cells to shear flow. Ras activation following 2 sec stimulation with shear flow was significantly more robust in the presence of full-length filamin, but not the actin- binding domain alone. This effect was not due to altered adherence of cells to the substrate. These results suggest that filamin is likely involved in sensing and/or transmitting the mechanical stimulus, and future studies will focus on determining the molecular mechanism of filamin action in this context.

## 2. Characterizing GPCR-G Protein Interactions in Real Time Through BRET2 Imaging

A.F.M. Tariqul Islam, Jamison Takashima, & Pascale G. Charest

(All affiliated with the University of Arizona)

Directed cellular migration towards a chemical signal, or chemotaxis, is critical for many biological functions, including wound healing, cancer metastasis, and immune responses. In *Dictyostelium discoidium*, a powerful model for cellular migration, chemotactic responses to the chemoattractant cAMP are initiated at cAR1, a cAMP-binding GPCR. While cAR1 and its associated G-protein have been researched extensively, real-time receptor-G protein interactions have not yet been characterized in *Dictyostelium*. We are using Bioluminescence Resonance Energy Transfer (BRET), a method of monitoring protein interactions in real time, to characterize the interaction between cAR1 and its associated G protein. Our preliminary data suggests receptor-G protein pre-coupling prior to receptor activation and shows an increase in cAR1-G protein association following cAMP stimulation. In the future, we plan to compare these results to similar BRET experiments using phosphomutated cAR1 constructs in order to better characterize the role of cAR1 phosphorylation in the adaptation of cAMP-dependent signaling pathways.

## 3. One of two prokaryotic deoxycytidine triphosphate deaminases found in *Dictyostelium discoideum* is functional and influences development

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Deoxycytidine triphosphate (dCTP) deaminase, an enzyme in the pathway that converts dCTP into dTTP, is typically found only in gram-negative bacteria and some archaea. Two genes encoding predicted dCTP deaminases (*dcd1* and *dcd2*) are identified in the annotated genome of *Dictyostelium discoideum*. Phylogenetic analyses suggest that *dcd1* and *dcd2* entered the *D. discoideum* genome through two independent horizontal gene transfer events from two types of bacteria. Dcd1 is most closely related to the Dcd of cyanobacterium *Synechococcus* and of *E. coli*, while Dcd2 is most closely related to the Dcd of the *Candidatus Yanofskybacteria bacterium,* an unculturable bacterium found in an aquifer in Colorado. A *dcd*⁻ *E. coli* strain, transformed with *dcd1*, expressed full-length Dcd1 and had a shorter lag phase compared to un-transformed bacteria. These findings suggest that *dcd1* encodes a functionally active protein related to the *E. coli* Dcd. However, the same *dcd*⁻ *E. coli* transformed with *dcd2* grew poorly and expressed truncated Dcd2. Currently, we are studying whether the activities and location of Dcd1 are regulated by the cell cycle and during development. Preliminary analyses of axenically-grown *D. discoideum* cells with a blasticidin-disrupted *dcd1*⁻ showed no apparent differences in growth rate compared to wild type Ax2. The starvation-induced development of the presumptive *dcd1*⁻ cells was delayed by 3-4 hours, suggesting a role for the Dcd1 during multicellular aggregation but no apparent differences in fruiting body sizes were observed. Under construction is a GFP-tagged Dcd1 that will be used to determine the cellular location of the enzyme during vegetative growth and development.

## 4. SRCP1 utilizes functional amyloid to suppress polyglutamine aggregation.

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The polyglutamine (polyQ) diseases are a class of incurable neurodegenerative diseases caused by the expansion of a polyQ tract in specific proteins, resulting in protein aggregation. One potential therapeutic approach is the development of strategies to reduce polyQ aggregation. We and others have found that the model organism *Dictyostelium discoideum* naturally encodes long polyQ tracts but is resistant to polyQ aggregation. Work from our laboratory has identified a novel molecular chaperone, serine-rich chaperone protein 1 (SRCP1), that is both necessary for *D. discoideum's* resistance to polyQ aggregation and sufficient to impart resistance to other organisms. The molecular mechanisms SRCP1 utilizes to suppress polyQ aggregation are currently unknown. Here we show that SRPC1 utilizes a functional amyloid domain to suppress polyQ aggregation. Using *in vitro* polyQ aggregation assays we demonstrate that a synthetic peptide of the functional amyloid domain is sufficient to suppress polyQ aggregation. To further understand structural aspects that mediate SRCP1 function, we have generated a computational model of SRCP1. Importantly, SRCP1's C-terminal domain is predicted to form a β-hairpin structure consistent with it forming functional amyloid. Interestingly, SRCP1's serine-rich N-terminus is predicted to interact with SRCP1's functional amyloid domain. Because serine-rich regions of proteins can suppress protein aggregation, this led us to hypothesize that SRCP1's N- terminal domain may function to suppress amyloid formation by SRCP1's C-terminal domain. Consistent with this we find that mutation of serines in SRCP1's N-terminus results in the formation of SRCP1 aggregates in cells. Together our data are consistent with SRCP1 utilizing functional amyloid to suppress polyQ aggregation.

## 5. A VIT/vWFA-Domain Containing Homolog of a Cancer Suppressor Interacts with an SCF E3 Ubiquitin Ligase in *Dictyostelium*

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BMB, CCRC, CTEGD, UGA

The SCF (Skp1, Cullin, F-box protein) class of E3 ubiquitin ligase serves an essential role in regulating motility, morphogenesis and differentiation in *Dictyostelium*. This is accomplished by tagging specific substrates for proteasomal degradation with polyubiquitin chains. Previous research has shown that genetic manipulation of Skp1 levels, or its oxygen-dependent hydroxylation and glycosylation, and disruption of F-box proteins such as FbxA/ChtA and MEKKα, can hinder *Dictyostelium's* ability to successfully fruit. However, through which substrates the SCF controls fruiting and how they are differentially regulated during development is unknown. We have found substrate candidates for another F-box protein, FbxD, which may be essential for growth and development, based on proteomic analyses of reciprocal co-immunoprecipitations. These previously uncharacterized but highly conserved proteins, Vvw1 and Vvw2, contain an N-terminal vault protein inter-alpha-trypsin domain (VIT) followed by a von Willebrand factor A domain and C-terminal domains labeled D3 and D4. This domain organization matches that of the poorly understood human cancer suppressor BCSC-1/VWA5A, for which one of these, Vvw1, shares 37% identity. The homology is concentrated in the consecutive VIT/vWFA/D3 domains. Preliminary studies suggest that Vvw1 is necessary for spore differentiation after culmination in strain AX4. Further investigation into FbxD's role in Vvw1 homeostasis, and VWA1's role in development, could establish *Dictyostelium* as a useful model for investigating the VWA5A protein family in human pathology.

**6. *Burkholderia* symbionts of *Dictyostelium*: Symbiont genotype and environmental context determine infection patterns and host outcomes**

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Symbionts can impart novel functions to their hosts. Symbiont infections and the phenotypes they impart are shaped by environmental context. Phenotypes induced by symbiont acquisition are and alter the phenotype to their hosts. Novel *Burkholderia* bacterial species *B. agricolaris, B. hayleyella,* and *B. bonniea* are natural symbionts of *Dictyostelium discoideum*. Approximately one quarter of wild *Dictyostelium* isolates are infected with *Burkholderia* (with *B. agricolaris* infections being most prevalent). Under typical lab conditions, *Burkholderia* symbiont species give rise to different host outcomes: *B. bonniea* is commenalistic, *B. agricolaris* moderately pathogenic, and *B. hayleyella* most pathogenic. Interestingly, *Burkholderia* symbionts, although inedible to amoebae themselves, induce secondary food carriage in their hosts (referred to as “farming”). Food carriage can benefit hosts when dispersed to new environments as they can “reseed” their food source. Here, we show that all *Burkholderia* symbionts induce some level of extracellular food carriage in host sori. However, *B. agricolaris* induces the highest level of both intra- and extra-cellular food carriage, and only *B. agricolaris* hosts show fitness benefits in food scarce environments (Khojandi, Haselkorn, Eschbach, Naser, & DiSalvo, 2019) . Furthermore, the extent of intracellular co-infections between *B. agricolaris* and secondary species is dependent on the identity of the secondary bacteria. In addition to food context altering *Burkholderia* infection outcomes, we find that temperature alters final *B. agricolaris* infection titers. Future experiments aim to elucidate the underlying mechanisms of secondary bacterial carriage and of temperature-controlled infection modification.

Khojandi, N., Haselkorn, T. S., Eschbach, M. N., Naser, R. A., & DiSalvo, S. (2019). Intracellular *Burkholderia* Symbionts induce extracellular secondary infections; driving diverse host outcomes that vary by genotype and environment. *The ISME Journal*. https://doi.org/10.1038/s41396-019-0419-7

## 7. Adenylyl Cyclase A mRNA Localized at the Back of Cells Is Actively Translated in Live Chemotaxing *Dictyostelium*

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*Dictyostelium discoideum* cells transport adenylyl cyclase A (ACA)-containing vesicles to the back of polarized cells to relay exogenous cAMP signals during chemotaxis. Fluorescence *in situ* hybridization (FISH) experiments showed that ACA mRNA is also asymmetrically distributed at the back of polarized cells. By using the MS2 bacteriophage system, we now visualize the distribution of ACA mRNA in live chemotaxing cells. We found that the ACA mRNA localization is not dependent on the translation of the protein product and requires multiple cis-acting elements within the ACA-coding sequence. We show that ACA mRNA is associated with actively translating ribosomes and is transported along microtubules towards the back of cells. By monitoring the recovery of ACA–YFP after photobleaching, we observed that local translation of ACA–YFP occurs at the back of cells. These data represent a novel functional role for localized translation in the relay of chemotactic signals during chemotaxis.

**8. Characterization of Mfsd8 in *Dictyostelium discoideum***

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The neuronal ceroid lipofuscinoses (NCLs), commonly known as Batten disease, are a group of untreatable, neurodegenerative lysosomal storage disorders. Ceroid lipofuscinosis neuronal 7 (CLN7) disease is a subtype of NCL that is caused by mutations in the *MFSD8* (major facilitator superfamily domain-containing 8) gene. *MFSD8* encodes a transmembrane protein that is predicted to play a role in transporting small substrates across membranes. The social amoeba *Dictyostelium discoideum* has been used extensively as a model organism in neurological disease research and encodes a homolog of human MFSD8. In this study, we show that *Dictyostelium* Mfsd8 GFP-fusion proteins, like human MFSD8, localize to compartments of the endocytic pathway. *Dictyostelium mfsd8*$^{-}$ cells display increased rates of proliferation in liquid media and enhanced growth on bacteria. When starved, *mfsd8*$^{-}$ cells display delayed aggregation and form larger mounds. Together, these results suggest that Mfsd8 may play an important role in regulating *Dictyostelium* growth and development.

## 9. Interplay of actin foci and clathrin rich areas upon *Dictyostelium discoideum* adhesion.

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Regulation of adhesion and unbinding is vital for versatile amoeboid movement in complex environments as soil. *Dictyostelium Discoideum (D.d)* expresses actin rich areas, so called actin foci. They are involved in adhesion, force transmission and are often colocalized to clathrin rich areas, which are involved in endocytotic engulfment. Clathrin coated structures (CCS) have been recently shown - for mammalian cells – to form plaque like structures involved in adhesion process and sensing of substrate rigidity.

To investigate coupling of clathrin rich areas and actin foci, we assess adhesion properties of wild-type and clathrin light chain ko. (CLC-) cells with AFM- based Single Cell Force Spectroscopy (SCFS). As shown previously, the stochastic step like rupture events visible upon cell retraction from surface rely on collective unbinding of adhesion proteins in bond clusters; especially these signatures vary between wild-type and CLC- cells.

Metal-induced energy transfer (MIET) allows to measure changes in fluorescence lifetime of fluorophors of interest in close vicinity to metal surfaces. This allows a spatial z-resolution of 250nm and 3nm accuracy of height information. It enables us to determine the height of membrane and freshly polymerised actin relative to each other within actin foci and CCS in wild-type and clathrin knock out cells.

We thus hypothesize that upon adhesome evolution, involvement of CCS in the adhesion process of *D.d.* has been crucial and the interplay between clathrin and actin foci is an important contribution to the overall versatility of the *D.d.* adhesion.

## 10. The role of GPCR in KIF1A-mediated vesicle transport

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Long-range vesicle transport in eukaryotic cells take place along a microtubule network that is usually organized as a radial array emanating from the centrosome. Cytoplasmic dynein is primarily responsible for transport toward the microtubule minus end anchored at the centrosome, whereas plus-end-directed transport is driven by the kinesin superfamily. In the past decade, many members of the kinesin and dynein super-families have been implicated in vesicle and organelle movement. However, how is motor activity regulated and what are the cargoes and receptors for each motor protein remain unresolved. Here we intend to investigate how membrane receptors GPCRs transmit signals to KIF1A-mediated vesicle transport.

## 11. Elucidation of the Physiological Function of N-terminal Ubiquitination by Ube2w

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Conjugation of ubiquitin to proteins regulates virtually every cellular process. Ubiquitin is attached to its substrates via a cascade of three enzymes. First, E1 ubiquitin-activating enzyme activates ubiquitin in an ATP-dependent manner, forming a thioester bond between the C-terminus of ubiquitin and its active site. E1's transfer ubiquitin to an active site cysteine residue on an E2 ubiquitin-conjugating enzyme. E2's then cooperate with E3 ubiquitin-ligases to facilitate the attachment of ubiquitin to substrates. Typically, ubiquitin is attached to the $\varepsilon$-$NH_2$ of lysine residues, but it can also be attached to the $\alpha$-$NH_2$ of the N-terminus ($\alpha$-$NH_2$) of proteins. Ube2w is the only identified E2 that preferentially ubiquitinates the $\alpha$-$NH_2$ of substrates. Previously, Ube2w knockout mice were found to be susceptible to post-natal lethality and had defects in skin, immune and male reproductive systems, though the pathways behind these defects are undefined. Despite the multisystemic effects caused by altered Ube2w expression, little is known about the physiological function of Ube2w or $\alpha$-$NH_2$ ubiquitination. To begin to define the cellular pathways regulated by Ube2w, we utilized the model organism *Dictyostelium discoideum* to generate Ube2w knockouts (Ube2w$^{KO}$) and performed RNA-seq. The results suggest that Ube2w plays a role in regulating cytosolic and mitochondrial metabolic pathways. Ube2w$^{KO}$ *D. discoideum* also show marked delays in development and fail to form mature fruiting bodies. Together these results suggest a potential role for Ube2w in regulating cellular metabolism, leading to downstream effects on chemotaxis and cell migration.

## 12. Identification of *Legionella*-resistant Genes in *Dictyostelium*

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Studies in the model organism, *Dictyostelium discoideum*, have led to the identification of a variety of host genes that modulate intracellular growth of several intracellular pathogens, including *Legionella pneumophila* and *Mycobacterium marinum*. However, there is a dearth of information regarding host resistance to these infectious organisms. In this study, we identified several genes in *Dictyostelium* which confer resistance to infection caused by *L. pneumophila*. To identify resistance-associated genes, we developed a novel 'resistance assay' using *Legionella*-susceptible wild type or resistant Nramp1-expressing *Dictyostelium* strain. Upon incubation of amoebae with *Legionella* for 96 hours post-infection, infected Nramp1-expressing cells could be recovered in liquid culture medium, as opposed to wild type cells which showed negligible growth. Using this screening assay, we evaluated 18 defined, single gene deletion *Dictyostelium* mutants for resistance to *Legionella* infection. Of these, 6 mutants, encompassing defects in phagocytosis, vesicle trafficking or chemotaxis, showed varying degree of resistance to infection. Preliminary studies have indicated that Nramp1 expression in the resistant mutant strains confer increased host resistance to bacterial infection. Encouraged by these results, we are currently attempting to introduce different combinations of these resistance-associated mutations in *Dictyostelium* for identifying host mechanisms which confer resistance to *Legionella* and other intracellular bacterial pathogens.

## 13. The Cellular Localization and Lipid Binding Properties of CpnF in *Dictyostelium*

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Copines belong to a family of calcium-dependent, phospholipid binding proteins that are characterized by the presence of two C2 domains and an A domain. The exact function of copines is unknown, but the proteins are speculated to be involved in membrane trafficking and cell signaling. This study aims to understand the function of copine F (CpnF) in the model organism, *Dictyostelium discoideum.* The lipid binding characteristics of CpnF were studied by immunoprecipitation and lipid dot blots, and the intracellular localization of CpnF was investigated using confocal microscopy. GFP-tagged CpnF was shown to bind to phosphorylated forms of phosphatidylinositol in a calcium-dependent manner. The strongest binding of CpnF was to PI(4)P and PI(3)P), with less binding to PI(4,5)P and (PI(3,5)P. Although GFP-CpnF bound to lipid dots in a calcium-dependent manner, binding to native cell membranes by centrifugation occurred in the absence of calcium and actin filaments. In fixed cells, GFP-tagged CpnF was found in the nucleus, and associated with endolysosomal organelles and the plasma membrane. In live cells, GFP-CpnF was in the cytoplasm and nucleus, but translocated to and from the plasma membrane in response to cAMP stimulation. Given that CpnF bound to phosphorylated forms of PI, which are important in endocytic processes, and was associated with endosomes and lysosomes, our data suggests that CpnF may play a role in endocytosis. We are currently working on creating a *cpnF* knockout mutant in *Dictyostelium* to study this further.

## 14. Towards understanding how primitive phagocytes discriminate Gram-positive and – negative bacteria

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Human phagocytic cells use two types of receptors for defense against invading bacterial pathogens: one for detecting and chasing pathogens via chemotaxis and another for recognizing and eliminating them via phagocytosis. Detection and chasing are facilitated by G-protein- coupled-receptors that sense diffusible chemoattractants derived from bacteria. Recognition and elimination employ pattern-recognition receptors (PRRs), such as Toll-like receptors for recognizing microbial-associated molecular pattern (MAMPs), and phagocytic tyrosine-kinase receptors for bacterial surface-bound complements or immunoglobulins. The social amoeba *Dictyostelium discoideum* are professional phagocytes that chase bacteria via chemotaxis and consume both Gram (-) and Gram (+) bacteria as food through phagocytosis. Yet, this stereotypical phagocyte does not encode orthologs of either Toll-like receptors or tyrosine-kinase receptors. Recent studies from our lab found that this stereotypical phagocyte utilizes fAR1, a class C GPCR, to simultaneously detect bacterial secreted folate for chasing bacteria and microbial-associated molecular patterns (MAMPs), lipopolysaccharide (LPS), for engulfing and consuming Gram (-) bacteria. An earlier study showed that Dictyostelium cells lacking functional heterotrimeric G- proteins are defective in phagocytosis in both Gram (+) and Gram (-) bacteria.

We propose that other C-class GPCRs are involved for recognizing the outer-membrane components of Gram (+). There are 14 C-class GPCRs encoded in the genome of Dictyostelium, and we believe that they represent a new class of receptors recognizing MAMPs. We are in the process to knock-out each of C-class GPCRs using CRISPR-CAS9 technique. We are going to identify the receptors recognizing surface molecules of Gram (+) and Gram (-) bacteria.

**15. Analysis of a MAPK docking site in the Gα2 G protein in Dictyostelium**

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MAP kinase interactions have been reported for G protein alpha subunits in yeast (Fus3-Gpa1) and Dictyostelium (Erk2-G.4). Support for such interactions have included mutational analysis of MAPK docking motifs. Alteration of the yeast Gpa1 docking site interferes with chemotropic growth. The Dictyostelium G.2 subunit contains a putative MAPK docking motif near its amino terminus in a position analogous to that in the yeast Gpa1 subunit. Alteration of this motif in G.2 (G.2D-) does not impact G.2 function when the G.2 subunit is overexpressed from the act15 promoter in high-copy vectors as suggested by the ability of G.2D-to rescue aggregation and cAMP chemotaxis in g.2-cells. This overexpression of the G.2D-or the wild-type G.2 subunit delays or prevents developmental progression beyond aggregation in g.2-or wild-type cells suggesting excessive G.2 function can interfere with multicellular development. In lower-copy vectors only the wild-type G.2 subunit, not the G.2D-subunit, is sufficient for mediating the aggregation phase of development. The reduced G.2D-expression results in impaired cAMP chemotaxis and increased cell dispersal. The G.2D-cells display normal GtaC transcriptional factor shuttling upon cAMP stimulation. These results suggest that the G.2 MAPK docking motif is important for cAMP chemotaxis and aggregation.

## 16. Basic methods for transformation in Dictyostelium caveatum

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Dictyostelium caveatum has a unique feature that they eat other species of social amoeba, however D. caveatum does not cannibalize each other. But it is unclear how they avoid cannibalism, because D. caveatum is not a model organism and molecular genetic experiment method has not been developed. Therefore we study the experimental character of D. caveatum and make transformation and REMI mutagenesis available in D. caveatum.

In our study, first, incubation methods was established. Wild strain of social amoebae needs two factors to do transformation experiment. One is that cells growth on agar plates. Another is that cells not be starved and keep in growth phase in liquid medium. Second, new plasmid vectors have been made by modifying the promotor region to be matched with D. caveatum, and we tested whether transformation using electroporation success. In this results, RFP or antibiotic resistance (hygromycin) was expressed in transformed cells, and therefore transformation success was established. But there was a problem that all hygromycin resistance strains could not form the fruiting body on the hygromycin plate. One of the possible cause was the combination of D. caveatum's hygromycin resistance and hygromycin activity. To solve the problem, we changed antibiotic drug hygromycin to neomycin. Accordingly, a new plasmid vector which contain a neomycin resistance gene was made. By using this new vector, the transformants formed fruiting body on the neomycin plate. So the basic methods for REMI mutagenesis have been developed.

**17. Copine C null mutants have defects in cytokinesis and phagocytosis**

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Copines are ubiquitously expressed calcium-dependent phospholipid binding proteins found in all eukaryotic organisms. They are characterized by two C2 domains in their N-terminus, followed by an A domain in their C-terminus and are thought to be involved in key aspects of membrane trafficking and calcium-dependent signaling pathways. Using *Dictyostelium discoideum* as a model organism, which has six copine homologs (*cpnA*-*cpnF*), researchers have started to characterize this family of proteins with the majority of investigations thus far focused on CpnA. To gain a greater understanding of the role of copines in *Dictyostelium,* a *cpnC* knockout cell line was created in axenic NC4A2 cells. Here we describe the results of the preliminary investigation into the *cpnC*- phenotype. Namely, the *cpnC*- cells were found to have normal developmental timing compared to the parental NC4A2 cells, but faster growth rates. Furthermore, *cpnC*- cells were shown to exhibit a slight cytokinesis defect, as well as a defect in their ability to phagocytose FITC beads compared to the parental NC4A2 cells. The *cpnC*- cell phenotype is similar to the *cpnA*- cells phenotype in that *cpnA*- cells also have faster growth rates and a slight cytokinesis defect. However, *cpnA*- cells are different from *cpnC*- cells in that *cpnA*- cells exhibit developmental defects and have increased rates of bead phagocytosis. A *cpnE* knockout cell line is being created and the resulting phenotype will be characterized and compared to that of the *cpnA*- and *cpnC*- cells.

## 18. Role of *PEPC* gene in prespore lineage choice and differentiation in *D. discoideum*

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During development of multicellular organisms, it is still unclear how cell lineages are primed and their own cell fates are retained towards final differentiation. To address these questions, we used *D. discoideum* to know how undifferentiated cells choose and thereafter maintain their own cell fate(prestalk/prespore) during development. Among a number of REMI mutants, we identified a mutation in the gene encoding phosphoenolpyruvate carboxylase (*pepc*) that functions in glycolysis. The *pepc-* mutant cells aggregated normally, but later exhibited a typical "slugger" phenotype. In the *pepc-* mutant slugs, expressions of a prespore-specific marker gene (*pspA*p:RFP) and prespore-specific antigen (PSV) are abolished, suggesting that *pepc* contributes to prespore cell differentiation.

Expression of a reporter gene driven by *pepc* promoter (*pepc*:GFP) was observed in prestalk region and more strongly in prespore region in wild-type slugs. Interestingly, heterogeneous expression of *pepc*:GFP was observed in a clonal cell population during the vegetative stage.

To investigate the relation between heterogeneous expression of *pepc* in vegetative cells and prespore-biased expression in development, we performed the Halo-tag pulse chase assay by labeling vegetative cells highly expressing *pepc* to trace their lineage. Those labelled cells were distributed in whole of the slug, but showed spore-biased differentiation. These results suggest that *pepc* contributes to the heterogeneity of growing cells for cell lineage choice and has a role in cell type specification into prespore/spore cell during multicellular development.

## 19. The Contractile Vacuole Localizes to the Rear of Migrating Dictyostelium and is Critical for cAMP Signal Relay

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Distinct changes often occur in morphology during the formation of cellular polarity. In the case of directed migration, cells often form a defined front and rear, as they migrate towards or away from migratory cues. In many cell types, the microtubule organizing center and nucleus take up discrete localizations relative to one another. During the directed migration that takes place in cellular aggregation, the early process in fruiting body development, *Dictyostelium discoideum* cells reorganize many cellular components, including their actin and microtubule cytoskeletons. We have discovered that the contractile vacuole (CV) network also polarizes towards the rear of the cell during migration. Observations with light microscopy of retrograde CV movement were confirmed by imaging cells expressing the fluorescently tagged CV marker dajumin. Interestingly, we found that the cAMP transporter ABCC8, which has been shown to be responsible for cAMP secretion, localizes to the CV network. Mutants lacking the huntingtin protein (*htt*$^-$) lacked a detectable CV in the vegetative state and had smaller than averaged sized CVs. *htt*$^-$ cells made very weak cAMP waves and didn't stream. *htt*$^-$ cells have previously been shown to have an inability to regulate their osmotic pressure, and the loss of this dynamic organelle is likely critical to this phenotype. Our results demonstrate that posterior redistribution of the contractile vacuole in migrating cells plays a critical role in recruiting cells into streams and contributes to cAMP secretion/relay.

## 20. Characterizing the mitochondrial proteome of *Dictyostelium discoideum* using quantitative mass spectroscopy for use in comparative proteomics

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Mutations on the mitochondrial genome (mtDNA) have emerged as important factors compromising human health. However, progress in understanding mtDNA genetics and diseases has been impeded by a lack of genetic approaches for studying mtDNA. Currently, there is no method to transform mitochondria in animal cells. The major hurdle towards successful mitochondrial transformation is to effectively deliver nucleic acids into the mitochondrial matrix. Curiously, mitochondrial transformation has been achieved in *Dictyostelium discoideum* using a routine electroporation procedure, suggesting that *D. discoideum* mitochondria are naturally competent. Consistent with this notion, *D. discoideum* does not possess a full suite of mitochondrial tRNAs on mtDNA and must transport nuclear-encoded tRNAs into the mitochondria to translate mtDNA-encoded proteins, underscoring the presence of machinery for the import of nucleic acids on *D. discoideum* mitochondria. To better understand the mitochondria tRNA importing process, we aim to use proteomic and genomic approaches, to identify proteins factors required for tRNA import and imported tRNAs molecules respectively in *D. discoideum.* Here we present our initial work characterizing the mitochondrial proteome of *D. discoideum* using quantitative mass spectroscopy. This work is part of a larger study to characterize the mechanism in *D. discoideum* responsible for nucleic acid import into the mitochondria with a long-term aspiration of transplanting a minimal system of mitochondrial nucleic acid import into other model organisms and enabling mitochondrial transfection in animal cells.

## 21. Copine A Null Mutants Exhibit Increased Phagocytosis and Adhesion

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Copines are highly conserved, calcium-dependent membrane binding proteins that have two C2 domains. *Dictyostelium discoideum* contains six copine genes, *cpnA-cpnF*, and our previous work on CpnA showed that CpnA associates with phagosomes. Here we investigated how CpnA is involved in phagocytosis and cell surface adhesion. By shaking cells on small petri dishes, we observed that *cpnA*- cells have increased adhesion to surfaces compared to parental cells. Using microscopy and flow cytometry we found that *cpnA*- cells adhered more 1 µm FITC beads and GFP-bacteria than their parental cell lines and this most likely caused the observed faster rate of phagocytosis of *cpnA*- cells. Using Latrunculin A to depolymerize actin filaments and inhibit phagocytosis, we found that more beads were associated with the surface of *cpnA*- cells than parental cells, suggesting the increased bead adherence is actin-independent. Bead adhesion assays in the presence of a calcium chelator indicated that the increased adhesion was not due to calcium-dependent adhesion proteins. Higher salt concentrations were needed to remove beads from *cpnA*- cells than parental cells, indicating this increased adhesion might be due to a change in electrostatic charge at the cell surface. Lastly, we used proteinase K to degrade proteins on the cell surface and found that while bead adhesion of parental cells decreased, more beads adhered to the surface of *cpnA*- cells, indicating proteins were not involved in the increased adhesion of *cpnA*- cells.

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